



National Health Service Standards 2nd Edition for Papua New Guinea

**Volume One
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National Department of Health

National Health Services Standards 2nd Edition Volume One

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NATIONAL HEALTH SERVICE STANDARDS

Applicable to: All Staff

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2011	Initial National Health Services Standards Vol 1-2-3	V1
2021	Modified - NHSS 2 nd Edition Vol 1-2	V2

FOREWORD

The 'National Health Service Standards' have significant impact for health service planning and resourcing for all health services in Papua New Guinea. The second edition of these Standards is aligned with the National Health Plan 2021-2030 (NHP), the governing policy document for the PNG health sector. The National Health Administration Act (NHAA) mandates the development of the National Health Plan, which sets out the goals and objectives to improve the standard of health of Papua New Guinea's people.

This 2nd edition of the National Health Services Standards (NHSS) aims to strengthen relationships between the achievement of the Sustainable Development Goals (SDGs) and Universal Health Coverage (UHC). Universal health coverage (UHC) is at the centre of current efforts to strengthen PNG's health systems and improve the level and distribution of health programs and health services. These standards translate the essential principles and values of UHC into a model of care, addressing access, coverage, and integration of health services.

The National Health Services Standards (2nd edition) are an important tool for providing direction and guidance to national, province and district planning and delivery of safe quality health services, and for informing clients, communities and stakeholders of the health services which they can expect to be available at each of the various levels of service delivery. Health Sector partners and stakeholders are required to work collaboratively to achieve these standards and implement public health and clinical care evidenced based interventions that align to these standards, our global obligations and relevant public health and clinical health policies and programs.

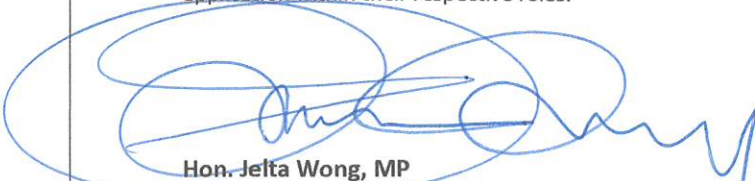
The National Health Service standards are essential to ensure that all health services are integrated and clearly articulated and can be planned, managed, monitored and evaluated in a strategic manner. They allow for health services planning at all levels and across all health programs and service types, inform financial, workforce, information and infrastructure requirements, clarify governance arrangements, and assist with evaluating ongoing care to ensure safe, quality health care delivery to meet optimal outcomes.

The National Health Service Standards outline the minimum requirements to be met to ensure safe, appropriate and quality health services delivery. These National Health Service Standards have three major purposes:

- i. To provide a standard set of criteria for 'Levels of Care', and 'Health Services' supported by the core components for health system strengthening that is accessible, equitable, affordable efficient, clinically viable and integrated with evidenced based public health interventions; and
- ii. To provide a quality improvement program with a set of National Quality Standards and an Accreditation process to guide health services through self-assessment, periodic review and a 4 year cycle of compulsory accreditation to meet Papua New Guinea National Quality Standards.
- iii. To provide health infrastructure and medical technology standards.

The PNG National Quality Standards (NQS) have been updated to ensure that the National Referral Specialist Tertiary and Teaching Hospital and Provincial Health Authorities (PHAs) have an integrated clinical and public health approach, supported by efficient corporate services support, to ensure quality and safety and in line with NHP KRAs. It is expected that all health services will strive to achieve these National Health Service Standards.

I commend these 2nd Edition National Health Service Standards, 2nd Edition to all PHAs, the National Referral Specialist Tertiary and Teaching Hospital and health partners, stakeholders and all health services staff for application within their respective roles.


Hon. Jelta Wong, MP
Minister for Health and HIV/AIDS



ACKNOWLEDGEMENT

The second edition of the **National Health Service Standards (NHSS)** set out a structure for the National Referral Specialist Tertiary and Teaching Hospital and Provincial Health Authorities (PHAs) to guide the delivery of safe, quality healthcare in Papua New Guinea, building upon the last 10 years implementation of the Standards. These Standards are important tool for strategic planning and assist health services to develop and/or review their 5 year Strategic Health Services Development Plans.

The National Health Service Standards when applied to health services delivery safeguard patient safety, increase community confidence in healthcare provision and improve health service quality and health facility design. The roles and functions of health services across PNG are governed by these National Health Service Standards. The Standards also provide the National Department of Health (NDoH) with the evidence to frame evidence-based budgeting for the health sector in Papua New Guinea. This ensures that we are able to fund all health services based on their correct levels of functional responsibilities.

Universal Health Coverage (UHC) underpins the need for consistent service standards approach across the whole of the PNG health system, and without this, there is a risk that pursuing individual targets separately will lead to even greater fragmentation of the PNG health system. The National Health Plan (NHP) 2021-2030 reinforces the need to ensure integration of all health services and programs, under-pinned by a more sustainable approach to the achievement of national health targets and creating a balance among them. The Key Result Areas (KRAs) of the NHP address: *engaged communities, working together in partnerships, universal health coverage, targeted priorities and strengthened health systems*.

To achieve UHC and meet the NHP KRAs for health services in PNG, the recent review of the National Health Standards (NHSS) refocused the Role Delineation for Health Services, strengthened the need for an integrated approach to 'Levels of Care and Health Services' and addressed targeted national health sector-wide priorities to include: healthy lifestyles and life stages, reduce the burden of infectious and communicable diseases and non-communicable diseases, and preparedness for disease outbreaks and emerging population threats.

The review also addressed NHSS Volume Two: Quality Standards, and the need for National Quality Standards and Compulsory Accreditation, based on a Package of Integrated Health Services Standards for the National Referral Specialist Tertiary and Teaching Hospital and Provincial Health Authorities (PHAs). The National Quality Standards strengthen quality improvements in healthcare and commitment to ensuring these National Health Services Standards (NHSS) are current, comprehensive and in alignment with the National Health Plan, best practice, and furthermore, that the standards are able to be assessed in a way that generates useful, accurate information for all health staff to improve safety and quality for all our patients and the broader community. We are committed to improving the quality and safety of our health care system. It is important that these National Health Service Standards are applied throughout the length and breadth of our country.

Health Boards of Governance, their Chief Executive Officers and Senior Management teams and their health partners are all held accountable for implementing these '**National Health Service Standards**', that their health services work towards meeting these minimum standards, and they undertake organisation-wide surveys and achieve accreditation status. Political leaders and the bureaucracy are asked to also apply these Standards when making decisions on health services infrastructure, equipment, supplies and human resources.

It is important to acknowledge the work and commitment by the National Department of Health Medical Standards Division Executive Manager and his team, the numerous senior clinicians, the contribution of members of public health for their ongoing advice and guidance and the assistance Health Services Sector Development Program (HSSDP) in the development of the second edition of National Health Service Standards.



Dr Osborne Liko
Secretary of Health

ACRONYMS

Acronym	Full terminology
AMS	Area Medical Store
HRH	Human Resources for Health
HTRC	Health Training and Resource Centre
ICT	Information and Communications Technology
KRA	Key Result Area
NCD	Non-Communicable Disease
NDOH	National Department of Health
NGO	Non-Government Organisation
NHAA	National Health Administration Act
NHB	National Health Board
NHIS	National Health Information System
NHP	National Health Plan
NHSS	National Health Service Standards
NQS	National Quality Standards
PFM	Public Financial Management
PHA	Provincial Health Authority
NRSTTH	National Referral Specialist Tertiary and Teaching Hospital and
UHC	Universal Health Coverage
SDG	Sustainable Development Goal
WISN	Workload Indicators of Staffing Need

EXECUTIVE SUMMARY

The National Health Administration Act (NHAA) mandates the development of the National Health Plan (NHP), which sets out the goals and objectives to improve the standard of health of Papua New Guinea's people.

Section 5 (1) of the NHAA says that the National Health Board can issue National Health Standards to implement the National Health Plan, on the recommendation of the Secretary of Health.

These Standards are the second edition endorsed by the National Health Board.

A formal review of the National Health Service Standards (NHSS) was overseen by the National Department of Health (NDOH) Medical Standards Division Executive Manager / Chief Medical Officer.

The revised NHSS, Second Edition addresses:

- a. reviewed levels of service from seven (7) levels of six (6) levels, including clinical patient care, appropriate public health interventions, and the national maternal health strategy;
- b. reviewed essential health workforce requirements, cadres of workforce and skill mix for primary and secondary health care at each level of care including utilisation of the health workforce productivity measures (WISN);
- c. reviewed health infrastructure standards to incorporate health facility planning design, medical technology, commissioning and facility management;
- d. a balance between rural and urban priorities; and
- e. inclusion of clinical & public health governance strategies.

There is a broader scope for services and health facility planning, management and monitoring of the implementation of minimum standards for supply chain/drugs and supplies/medical equipment/devices and consumables.

The review of the NHSS considered the efficiency factors linked to financial management & activity-based funding models and a compulsory Provincial Health Authorities (PHA) and the National Referral Specialist Tertiary and Teaching Hospital accreditation process of health services delivery, including that of health partners.

The National Health Service Standards outline the minimum requirements to be met to ensure safe, appropriate and quality health services delivery.

These National Health Service Standards have three major purposes:

- i. To provide a standard set of criteria for 'Levels of Care' and 'Health Services' supported by the core components for health system strengthening that are accessible, equitable,

affordable efficient, clinically viable and integrated with evidenced based public health interventions;

- ii. To provide a quality improvement program supported by a set of National Quality Standards and an Accreditation process to guide health services through self-assessment, periodic review and a 4 year cycle of compulsory accreditation to meet Papua New Guinea NHSS National Quality Standards;
- iii. To provide a set of standards that guide health facilities infrastructure and medical technology planning, design, development, commissioning, facility management and assets disposal that is informed by clinical and public health services requirements of the six (6) levels of care.

The review of NHSS Volume One, strengthens the need for all health sector partners and stakeholders to work collaboratively with the National Referral Specialist Tertiary and Teaching Hospital and PHAs to achieve these standards and implement integrated public health and clinical care evidenced-based interventions, that align to standards, global obligations and relevant public health and clinical health policies and programs. These standards will also apply to all health sector stakeholders including church health service providers, private sector health providers and non-government organisations (NGOs) providing health services across PHAs and with and the National Referral Specialist Tertiary and Teaching Hospital (NRSTTH).

The review of NHSS Volume Two: Quality Standards, addresses the broader National Quality Standards and Compulsory Accreditation based on a Package of Integrated Health Services Standards for Provincial Health Authorities (PHAs) and NRSTTH.

The National Quality Standards (NQS) strengthens quality improvements in healthcare and commitment to ensuring these National Health Services Standards (NHSS) are current, comprehensive and in alignment with the National Health Plan, best practice, and furthermore, that the standards are able to be assessed in a way that generates useful, accurate information for all health staff to improve safety and quality for all patients and the broader community. It is important that these National Health Service Standards are applied throughout the length and breadth of the country.

Building on the foundation of the 2011 National Health Service Standards, these revised 2021 NHSS continue to provide a rational set of standards for allocating scarce financial, physical and human resources between those various levels of the hierarchy (levels of care) to encourage the development of viable clinical and public health services in the most efficient and effective manner possible.

The National Health Services Standards provide a strategic overview for public health, primary health services and clinical care and enable 5-year strategic health services development planning.

The standards are essential to ensure that all health services and programs are integrated and clearly articulated and can be planned in a strategic manner. They account for workforce planning; inform financial and infrastructure planning; clarify governance arrangements, support a formalised

patient referral pathway, and evaluate care to ensure safe, quality health care is being provided to deliver optimal outcomes.

BACKGROUND

The NHAA 1997 requires the National Department of Health (NDOH) to develop a single health policy (the National Health Plan 2021-2030) and National Health Standards. Within the National Department of Health's Medical Standards Division and in partnership with the Public Health Division, both divisions are responsible for the development, monitoring and evaluating of health care standards in Papua New Guinea, and reporting on governance reform programs to improve overall health services delivery and health outcomes.

To fulfil the NHAA 1997 mandate the **2021-2030 National Health Service Standards (NHSS)** are developed to outline:

- A model of care that supports the National Health Plan 2021-2030 Key Result Areas (KRAs);
- Levels of Care and Health Services for all Provincial Health Authority (PHA) health services delivery in Papua New Guinea;
- National Quality Standards which lead to a formal process of self-assessment, periodic review, NRSTH and PHA organisation-wide accreditation survey by peer surveyors, a program of continuous quality improvement and a status of accreditation obtained by a PHA-NRSTH after a successful external survey evaluation and
- Health infrastructure and medical technology standards.

NATIONAL HEALTH SERVICE STANDARDS FOR PAPUA NEW GUINEA

The NHSS provide a nationally consistent statement of the level of care and health services patients/consumers can expect from NRSTTH - PHAs and other health service organisations, and aims to ensure that the PNG health system is better informed, supported, organised and integrated to deliver safe and high-quality health care.

Individual PHAs together with their health partners, can define their health service delivery models based on the NHP, 5-year strategic health service development plan, and strengthen governance systems and processes and formalise patient referral pathways in line with their health service delivery models.

The review of National Health Plan 2011 to 2020 and the subsequent development of the National Health Plan 2021 to 2030 followed consultation meetings with the NRSTTH, PHAs, Health Partners and other key stakeholders, and provided policy decisions on the levels of health service facilities particularly the aid post and regional hospital levels.

The focus was on rationalising levels of health services and moving forward, addressing accessible, equitable, affordable, integrated, evidenced-based essential health services; public health interventions, primary health and clinical care and services working together; in consideration of the social and economic determinants of health and prioritisation of the known burdens of disease in Papua New Guinea.

The **Papua New Guinea National Health Plan 2021-2030** envisages the following Six (6) Level hierarchical structure for PNG health services across the nation, in line with a revised role delineation:

- **Level.1 (Village) Health Posts (HPs)** – basic-level providing primary health care including health promotion, health improvement, health protection.
- **Level.2 Community Health Posts (CHPs) and Health Services** - primary health, ambulatory care and short stay inpatient and maternity care at the local rural / remote community level, with a minimum of six (6) health workers to ensure safe 24 hour care and treatment
- **Level.3 Health Centres / Urban Clinics** –offering primary health and ambulatory care in urban and rural settings and inpatient maternity and newborn care in major provincial urban communities
- **Level.4 District Hospitals and Rural Health Services** - offering primary and secondary level clinical services and district wide public health programs
- **Level.5 PHA Provincial Hospitals, Health Services and Public Health Programs** – offering secondary level & specialist clinical care services and supporting primary health care and integrating public health programs and patient referral
- **Level.6 National Referral Specialist Tertiary and Teaching Hospital and Health Services** – offering complex tertiary level clinical services and supporting primary health care and public health programs and a formalised patient referral arrangement.

A MODEL OF CARE FOR HEALTH SERVICES FOR PAPUA NEW GUINEA

The 2021-2030 National Health Plan's vision is building the health of our people, leaving no one behind can be enjoyed by all. The goal is preventing ill health, identifying and addressing health risks and emergent diseases and providing accessible, affordable, safe, quality health care to all. The mission requires that Communities, Government and Health Partners all work together to promote health and well-being and deliver compassionate and equitable health care for all.

The essential principles and values of the PNG health system are:

- **People Centred:** Services that are focused on people, family, environment and culture that are comprehensive, accessible and community-based that meet health needs of people throughout their life course.
- **Equity:** Services that are equitably distributed that reach all people regardless of their sex, age, disability, political affiliation, religion, culture, gender or ethnicity.
- **Partnerships:** Working with all partners across all levels of the health system.
- **Health services adapted to community needs:** Accessible to safe, quality and affordable health care that is responsive to specific needs.
- **Integrated Approaches:** Balanced interventions addressing primary, secondary and tertiary levels of health care and health programs within the PNG health system.
- **Evidence-Based Innovation:** Pursuing high quality and safe health outcomes through research, science and technology that meets international standards.

All Provincial health Authorities (PHAs) and National Referral Specialist Tertiary and Teaching Hospital develop a model of care that reflects the essential NHP principles and values, and support the range of essential clinical, primary health care and public health interventions and services.

The model of care should focus on continuously improving patient care throughout the health system, ensuring that its coverage extends from self-care management and engaging with communities, disease and injury prevention and health promotion, early detection and intervention, to integration and continuity of health care.

The aim of a model of care is for all individuals and communities receive the health services they need without suffering financial hardship and includes the spectrum of safe, quality health services, from health promotion to prevention, treatment, rehabilitation, and palliative care across the course of individual's life.

The delivery of these health services requires strengthening PNG PHA and National Referral Specialist Tertiary and Teaching Hospital health systems to include:

- Emphasizing not only what services are provided, but how they are funded, managed, and delivered;
- Improving the focus of health promotion and disease prevention at community-village level and each level of health service delivery;
- Improving health service coverage and health outcomes, in terms of availability and accessibility;
- Improving robust budgeting, financing and accounting structures;
- Improving procurement and medical supply chain processes;
- improving the capacity and capabilities of all health staff and support workers to deliver safe, quality, people-centred and integrated health services;
- Improving health infrastructures, associated health staff accommodations, together with their preventative maintenance;
- Improving availability of health assets, equipment's, medical technology, transport, and logistics; and
- Improving abilities to respond to emergencies and disasters, and manage risks more effectively.

A fundamental shift in health services delivery is needed, such that services are integrated and focused on the needs of people and communities. This will include reorienting National Referral Specialist Tertiary and Teaching Hospital (NRSTTH), PHA health services and health programs to ensure that care is provided in the most appropriate setting, with the right balance between ambulatory care and inpatient care and strengthening the coordination of care and programs. Ensuring that health services are organized around the comprehensive needs and expectations of people and communities helps empower them to take a more active role in their own health and the health system.

Primary Health Care (PHC) entails three inter-related and synergistic components, including:

- comprehensive integrated health services that embrace primary care as well as public health goods and functions as its centre-pieces;
- multi-sectoral policies and actions to address the upstream and wider determinants of health; and
- engaging and empowering individuals, families, and communities for increased social participation and enhanced self-care and self-reliance in one's own health.

Investments in quality primary health care is the cornerstone.

The primary health care approach and life course (stages) approaches are critical. A primary health care approach focuses on organizing and strengthening health systems so that people can access

services for their health and wellbeing based on their needs, at the earliest, and in their everyday environments.

There must be an integrated approach and balanced interventions that address primary, secondary and tertiary levels of care within the overall health care system.

There are significant challenges that must be met to ensure that health care in Papua New Guinea is provided and continuously improved to enable safe and quality care. These challenges include:

- increasing patient-consumer expectations;
- expensive technologies & new treatment innovations;
- a growing burden of chronic & disabling conditions (infectious & non communicable).;
- poor patient referral pathways, little continuity and/or coordinated care;
- chronic under-funding and under-staffing of the health system; and
- lack of effective monitoring and evaluating of the performance of health services.

Church-run health facilities are integral to PNG's health system and estimated to provide almost half (50%) of PHC services across the country. Health facilities managed by churches currently operate somewhat independently from government-run facilities, nevertheless, are highly subsidised, as approx. 80% of their operational costs are financed by the PNG government.

The existing PNG health system is stretched beyond capacity in response to coronavirus disease of 2019 (COVID-19) pandemic.

District Development Authorities (DDAs) - Open MPs (eighty-nine [89] single-member districts) play an increasingly significant role in financing health infrastructure and/or future incentive projects.

The PNG government struggles to translate its investments in health systems into improved performance on the ground and a World Bank review on rural health financing found that provincial governments increased spending on health services but did not show improvement against key health indicators, as confirmed via data collected from eNHIS.

The NHSS Volume Two: National Quality Standards (NQS) reflect the importance of integration within a model of care, with the main points being:

- PHAs prioritize integrated outreach to hard-to-reach communities;
- NDOH, National Referral Specialist Tertiary and Teaching Hospital (NRSTTH), and PHAs work collaboratively with Health Partners and other key stakeholders to deliver an integrated and inclusive approach that is responsive to communities needs and priorities;
- PHAs guide Government agencies, including District Development Authorities (DDAs), to establish and deliver integrated community focused health services at district and LLG community/village levels;
- NHSS include six levels with comprehensive health promotion and prevention integrated at all levels from Level 1 – (Village) Health Post through to Level 6 – National Referral Specialist Tertiary and Teaching Hospital;
- NDOH, National Referral Specialist Tertiary and Teaching Hospital and PHAs improve integrated health sector planning, budgeting and monitoring to enhance effective use of funds; and
- NDOH, National Referral Specialist Tertiary and Teaching Hospital and PHAs work in collaboration with relevant GoPNG agencies implement a national health sector wide area network that connects

all provincial capitals and Port Moresby to provide integrated ICT communication and data services by 2030.

The **benefits of the National Health Service Standards** include:

- Improved balance of preventive, primary and acute (curative) care and public health interventions;
- Improved National Referral Specialist Tertiary and Teaching Hospital and PHA models of care with greater emphasis on client-patient focus and a strengthening of primary health care for all;
- Comprehensive definitions of essential health services for each health service (acute-curative care, public and primary health care) and relationships between health services;
- Capacity planning for geographical areas and catchment populations;
- Investment in health service infrastructure and equipment to support service delivery;
- Investment in health workforce and skill sets to deliver health services in accordance with levels of care;
- Continuous professional development, education and upskilling of health professionals and staff development requirements which support maintenance of a skilled and dedicated health workforce for each level of health service delivery;
- Encourage service planning and consider risk management procedures when services do not meet minimum requirements for patient and community safety;
- Improve access to appropriate health care services;
- Utilise national standards, as a formal quality improvement process to ensure the delivery of safe, quality health care through accreditation; and
- Invest in a continuous improvement program where health care managers, providers and health departments can ask 'are we doing things right' and 'are we doing the right things' to improve safety and quality of patient care and wellbeing of communities as well as identify key areas of risk and adverse events which are managed, monitored and reported.

LEVELS OF CARE AND HEALTH SERVICES IN PAPUA NEW GUINEA

An effective health system is one that meets Universal Health Coverage (UHC) objectives by providing equitable access to affordable, high-quality health care—including treatment and curative services, health promotion, prevention, and rehabilitation services—to the entire population.

Unfortunately, PNG's healthcare system does not currently meet these standards.

Shortfalls in access, quality, safety, efficiency, and equity have all been well documented. Public health and clinical care are now required to focus on interventions that are the most cost-effective and improve the health of the worst-off populations.

The 2011 NHSS Role Delineation Matrix, which identified the role delineation of health services and service level definitions, has been reviewed, and a new set of prioritised health services covering six (6) levels of care linked to health systems strengthening and proven interventions for their effectiveness or cost effectiveness is updated.

The NDoH Senior Executive Management Team approved the use of the revised six (6) levels of health facilities and health services that are outlined in this 2nd edition NHSS, and in the NHP 2021 to 2030 is structured accordingly to reflect these revised levels.

The levels are now as follows:

• Level 1: (Village) Health Post: is maintained as the most rural and remote peripheral health service facility.
• Level 2: Community Health Post and health service: combines existing Health Sub Centre and Community Health Posts.
• Level 3: Health Centre and health service: are for Rural areas and Urban settings with Urban Clinics as day only ambulatory service for urban populations.
• Level 4: District Hospital and rural health service: remain as a designated PHA district hospital supported and integrated with public health programs.
• Level 5: Provincial Hospital: remain as the PHA provincial general and specialist referral hospital supported and integrated with public health programs. Provincial Health Authorities to build their specialist hospital capacities and capabilities.
• Level 6: National Referral Specialist Tertiary and Teaching Hospital of PNG: previously the Port Moresby General Hospital (PMGH).

Noting, the former categories of Regional Hospitals cease to exist with 'sub specialist' referral functions managed by the National Referral Specialist Tertiary and Teaching Hospital PMGH, and Health Sub-Centres being incorporated into Community Health Posts.

Health care services and health facilities are to ensure safe quality patient care and create health services that invest in the population's wellbeing and enable timely access to appropriate health services.

This focuses on utilising primary health services as the initial point of access and facilitate health promotion, preventative health and health improvement, as these are an integral component of clinical patient, community care and public health interventions.

National Referral Specialist Tertiary and Teaching Hospital and Provincial Health Authorities (PHAs) Levels of Care are based on health services and health programs that are evidenced-based, viable interventions and integrated public health strategies at each level of care, they include:

Level	Levels of Service	Broad definition
Level 1	Health Post (HP) and preventative/promotion services	Set within communities-villages: to strengthen public health, primary health care; village level engaged communities: - ensuring greater recognition of the Primary Healthcare role of Village Health Assistants (VHAs); supported by Community Health Workers (CHWs); - matching Health (Aid) Posts within remote locations; - engaging more with Non-Government Organisations / Churches and LLG - DDAs for community-based programs; - reflecting innovative roles of VHAs: defining scope of practice, competencies centred on preventive and promotive activities; - supporting supervision and training, (example for Certificate II qualification registration-licencing of VHAs in collaboration with regulatory bodies i.e. Medical Board & Nursing Council, and - delineating clearly, accountability and funding streams, partnering with LLGs and DDAs.
Level 2	Community Health Post (CHP) and Health Services	Community Health Posts (CHP) is the entry into the formal healthcare system. The existing CHP Policy / Guideline is incorporated within NHSS: - CHP is the 'Hub' for primary health care, a strong focus on health prevention and support of 'Village Health Assistants & existing Health Aid Posts within villages'; public health activities including outreach patrols, awareness campaigns and health promotion activities are delivered. - CHP providing integrated and more accessible services, defining mechanisms for supervision of VHAs, roles of CHWs i.e. competencies, practices and upskilling programs i.e. CHW Upskilling Program for Mothers & Newborn. - CHPs continuing to increase facility based primary health care activity and emergency services and recognising increased workforce productivity, essential skill mix and integration of public health programs; - CHPs prioritising access to respectful maternity care, which includes antenatal care, supervised deliveries and postnatal care aiming to improve maternal and newborn health outcomes; - CHP providing a significant role in the 'initial point of care, stabilising and resuscitation of patients and patient referral; and - CHP workforce establishments aligning with PHA priorities for the provision of good governance, staffing, safe quality twenty-four (24) hour clinical and primary health care and supportive supervision.
Level 3	Health Centre (HCs) and Urban Clinics (UCs) and Health Services	Patients are the centre of care; clinical interventions are based on evidence: - maintaining high standards of safety & quality; - increasing patient awareness and public health interventions; - improving early diagnosis and interventions with the aim of identifying proven efficiency and effectiveness; and

		including Urban Health Services in major provincial centres and Urban Day Clinics in major provincial and district centres.
Level 4	District Hospitals and Rural Health Services	The criteria for planning a District hospital and Rural health services includes: - defining populations across Districts i.e. catchment populations; - identifying access & referral pathways (roads, rivers); - prioritising local burden of diseases (data driven – evidenced based clinical practice & public health interventions); - defining health service functions and standards including quality & safety; - ensuring essential workforce planning –cadres – training – continuing professional development (in-service); and - integrating clinical services, public health interventions & primary health care. Many existing health facilities are unable to ensure these services are fully operational (equipment-drugs-supplies-operational budget) and staffed (skilled competent cadres of health workers), especially in newly built health facilities.
Level 5	Provincial Hospitals supported by Urban Health Services and Public Health Programs	Provide Specialised and Subspecialised health services: - ensuring coordination and integration of clinical services, public health interventions and primary health care; - ensuring patient/people centred, evidence, safe based on a culture of continuous improvement, proven efficiency and effectiveness; - accessing health information to support continuous quality improvements – Audits, Peer Review, etc. A menu of Sub Specialised healthcare is defined in the health services matrix (attached), following analyses of UHC relevant to Papua New Guinea; - setting priority is based on available resources e.g. funding, workforce, recurrent costing, etc. - financing options may include income generation, fee for service, schedule of fees and consideration given to the role of Private Health Sector, Public Private Partnerships and high-end technology costs i.e. Radiology / ICT /IT connectivity.
Level 6	National Referral Specialist Tertiary and Teaching Hospital and Public Health Programs	The National Referral Specialist Tertiary and Teaching Hospital provides tertiary care, a level of health care obtained from Specialists after referral from primary health care and secondary care: - ensuring a full complement of services including ‘high end’ Specialist Paediatrics, Obstetrics, Surgery, Medicine, Gynaecology, subspecialty Surgery for major operations, consultations with subspecialists and with sophisticated intensive care facilities and/or dedicated to Specific Sub-Specialty Care ie health services for Oncology as a Care Centre are defined at both National Referral and ANGAU hospitals - Psychiatric hospital remains at Loloki).

The existing role delineation has been further reshaped to **‘Levels of Care and Defining Health Services’** that are accessible, equitable, affordable efficient and clinically viable, integrated and address evidenced based public health interventions, UHC principles and targeted health priorities.

The health priorities include:

- healthy lifestyles,

- life stages,
- burden of infectious and communicable diseases and non-communicable diseases, and
- preparedness for disease outbreaks and emerging population threats.

These is reflected in the NHP KRAs, objectives and strategies and the National Quality Standards for health services in PNG. The 'Levels of Care and Health Services' Matrix focusing on an integration of health services addressing: community engagement, public health programs, inpatient (acute care) services, ambulatory care (i.e. outpatient) services, primary health care, disease prevention and promotion, clinical care, diagnostic services, allied health and support services and the NRSTTH, and PHA capabilities (health system strengthening).

PHA Hospitals and National Referral Specialist Tertiary and Teaching Hospital 'Levels of Care and Health Services' Matrix outlines the sub specialised services with a focus on access and coverage, public health and clinical functions 'range of services' which are coordinated and integrated, delivered effectively and efficiently, and supported by specialised clinical networks. There is evidence of coordination and integration of clinical services and linkages to public health evidence-based interventions; person-centred, based on evidence and within a culture of continuous improvement and efficiency and effectiveness in the provision of services.

The matrix shows the complexity of level and health service and reflected in 'clusters' or groups that are relevant for the catchment communities and targeted sub population groups; not all health facilities have all levels of care and all services. For example, the high-level tertiary hospital will be the site to deliver highly sub specialised level of care. The process defines health services and the six level at which these are provided. Its interpretation enables some degree of flexibility, combined with consideration for the six (6) functional levels.

The 'Levels of Care and Health Services' Matrix provides a consistent language to describe health services and acts as a tool for planning service development. The National Health Service Standards 'Levels of Care and Health Services' Matrix aims to improve the management of demand for healthcare services.

The Matrix does not attempt to describe all the services which are provided by all health services, but confines itself to those which are widely considered to be the prioritized essential services, evidenced based and cost effective. The matrix includes only brief descriptions of the capabilities and requirements of the service groups including workforce and medical equipment.

Health service planners, clinical and public health leaders can also consider a cluster of services across sites and programs that could be within a district catchment-based structure or a network of services rather than a facility focus. Templates are provided for the 'clusters of health services' enabling National Referral Specialist Tertiary and Teaching Hospital and PHA to benchmark against service capacity levels for each health facility or service within an PHA.

This is achieved through measuring the integration of health services across setting; planning evidenced based public health programs for targeted populations programs, utilising the assessments to improve patient flow, availability of beds, free up acute (curative) care capacity and

strengthening service delivery networks and networking. There is a need to consider together with patient referral patterns and patient flows, population clusters and transport infrastructure (road, sea and air). A service network or district health service may be required to achieve a functional health service across a set of health facilities and public health programs.

LEVELS OF CARE AND HEALTH SERVICES MATRIX

The Health Services Matrix is used to determine essential health services and levels of care in relation to a health service's everyday activity.

The service matrix should be used to:

- Match the services provided in/or across health services and mainstream targeted health promotion and education programs and activities for reducing the incidence and impact of health problem
- Provide consistent and comparable information on the level of care and service delivery including the type of service;
- Identify the relationship between the service groups;
- Enable strategic and operational planning based on consistent service description;
- Encourage service planners to review service profiles on a comprehensive scale, with consideration of a district wide or network of health services in a province; and
- Review services in response to demands and changes.

'Cluster' - Group Information is listed as follows:

HEALTHY LIFESTYLES: HEALTH PROMOTION AND EDUCATION

The WHO definition of a **Healthy Lifestyle** is: *'A complete state of mental, physical and social well-being not merely the absence of disease'*. A healthy lifestyle is generally characterized as a “balanced life” in which one makes “wise choices”. Healthy Living definition is the steps, actions and strategies one puts in place to achieve optimum health. Healthy Living is about taking responsibility for one's own decisions and making smart health choices for today and for the future.

A healthy lifestyle is a valuable resource for reducing the incidence and impact of health problems, for recovery, for coping with life stressors, and for improving the quality of life. There is a growing body of scientific evidence that shows lifestyles play a huge part in how healthy the community or population are. From what people eat and drink, to how much exercise they take, and whether they smoke or take drugs, all will affect their health, not only in terms of life expectancy, but how long they can expect to live without experiencing chronic disease.

Conditions such as heart disease, cancer, diabetes, joint disease, and mental illness are responsible for a vast number of deaths and disabilities. Healthcare services almost exclusively focus on the provision of clinical care by highly trained health professionals as the major strategy to deal with these conditions. Many health problems can be prevented or at least their occurrence postponed by having a healthy lifestyle.

The Vision for Health Promotion in PNG, is to empower individuals and communities thereby enabling them to control the status of their own health.

To realize this Vision for health promotion requires the participation of individuals, communities and political leaders at all levels. While it is the individual that ultimately chooses a healthy lifestyle, government and society as a whole have a responsibility to make healthy choices easy, early, exciting and everywhere by ensuring that supportive environments are available in various settings.

Health Promotion and prevention programs and activities are key components of any healthcare system and informs public health interventions to address public health issues through a coordinated and multisectoral approach.

All health workers from Village Health Posts, Community Health Posts, Health Centres, District and Provincial Hospitals and National Referral Hospital have basic knowledge, and adequate skills to plan and facilitate health awareness, health promotion, and education across levels of care and health service delivery.

The results of not implementing health promotion programs and activities may result in: an increasing burden in the trend and spread of infectious diseases; increase in the numbers and burden of non-communicable diseases; rise in neglected tropical diseases that were previous under control and now re-emerging; sporadic disease outbreaks of vaccine preventable childhood diseases e.g. polio and measles; and increasing number of trauma and injuries that are over burdening our health care facilities.

Health Promotion activities are conducted at every level of care and integrated into health services for patients and communities and supported by the following pillars:

- NDOH, National Referral Specialist Tertiary and Teaching Hospital and PHAs strengthen partnerships to extend and amplify the impact of health promotion; with great potential for collaboration with government agencies, NGOs, Churches and community groups and others with strategy that calls for a multi-sectoral and multi-disciplinary approaches to promote health in settings such as village, schools, health facilities, public transport, and restaurants, etc. addressing preventable and communicable and non-communicable diseases.
- NDOH supports health IEC productions for PHAs health promotion activities
- All PHA and National Referral Specialist Tertiary and Teaching Hospital have a network of skilled health workers with the capability for the promotion of 'healthy lifestyles supported by accurate, up-to-date information, tools and strategies.
- PHA, National Referral Specialist Tertiary and Teaching Hospital and health partners implement national health promotion programs via health promotion actions, ie education (through IEC materials); social mobilization through community participation); and advocacy (through use of mass media).

In order to increase the visibility and prominence of health promotion and education programs and activities as a core component of the primary health care the following key strategies are prioritized:

- Strengthen primary health care across PNG health systems as reflected in these NHSS
- National Referral Specialist Tertiary and Teaching Hospital and PHA structures reflect key health promotion education positions at National hospital, PHA provincial and district levels
- Evidence of targeted funding for health promotion and education programs at the National hospital, PHA provincial and district levels;
- Strengthen the leadership and management skills of health promotion practitioners of programs in the provinces and districts;
- Develop partnerships with other health and non-health partners in the provinces and districts to expand the roll out of health promotion and prevention programs as a core component of the primary health care.

Communication is an important part of health promotion and is challenging. An effective and sustainable health communication network that reaches communities in rural-remote and urban areas is appropriate. Communication strategies determine collectively through dialogue and debate with communities and partner organizations.

LIFE COURSE

A life course approach emphasises a social perspective, looking back across an individual's or a cohort's life experiences or across generations for clues to current patterns of health and disease, whilst recognising that both past and present experiences are shaped by the wider social, economic and cultural context.

The importance of the life course as a key principle has enormous implications on the way an individual's health is considered, for the delivery of healthcare, training of health professionals and for the way healthcare systems are developed to cater for individuals' health care needs.

A life course framework is being used to consider life stages (i.e. *gestation, childhood, adolescence, young adulthood and midlife*) that affect chronic disease risk and health outcomes and to assist in planning and priorities health services and includes:

- the importance of all ages and stages of life and recognises the dimension of health and ageing, rather than just distinct episodes of illness; as part of a life process;
- the opportunity to focus on 'health' as well as specific disease processes; in terms of disease, it emphasises health promotion, disease prevention and treatment and disease management throughout life;
- the emphasis on primary interventions in addition to treatment or palliation; and the requirement for an interdisciplinary and multidisciplinary approach to improve overall coordination of care, services and programs.
- The assurance that people's health problems are addressed through comprehensive promotive, protective, preventive, curative, rehabilitative and palliative care throughout the life course, prioritizing primary care and public health services as the central elements of integrated service delivery and include:
 - Reproductive Health & Contraception
 - Maternal & Newborn Health (Obstetric), (discharges)
 - Child Health, (nutrition), School Age Health & Development
 - Adolescent Health & Development
 - Women's and Men's Health, Care of Elders and Disabled

all integrated into levels of care for Communicable and Non-Communicable Disease health services.

Supporting health information:

Global Indicators:

- ▲ Reduce Under 5 mortality rates from 45/1000 to below 20
 - ▲ End preventable newborn & children under 5 yr: neonatal mortality as low as 12/1,000 live births & under 5 low as 25/1,000 live births
 - ▲ Reduce Maternal Mortality rate form 171/100,000 to below 100 (to less than 70/100,000 live births)
- Ensure universal access to sexual and reproductive health-care services, including for family planning, information and education, and the integration of reproductive health into national strategies and programs

PNG health information:

- I. *Early childhood mortality has decreased from 159 per 1,000 births in 1967 to 49 in 2018, however of the 313,000 births in 2020, over 15,000 will die before their 5th birthday with many dying as newborns.*
- II. *Maternal mortality has also declined from a high in 2006 of 733 per 100,000 to 171, yet this still means at least one woman dies every day somewhere in PNG due to delivery complications.*
- III. *Neonates (0 – 28 days of life) account for just over a third of all childhood deaths, mostly due to neonatal sepsis, birth asphyxia, and very low birth weight*
- IV. *10% of all paediatric hospital admissions are children who are severely malnourished.*
- V. *Some form of disability is likely to affect 1.3 – 1.4 million people in PNG.*

PNG Enhis Indicators:

- I. Early childhood mortality rates:
 - a. Neonatal mortality rate 20/1000 live births;
 - b. Infant Mortality Rate = 33/1000 live births;
 - c. Under-5 Mortality Rate = 49/1000 live births.
 - d. 1/5th children at MCH) clinics underweight: Stunting significant concern with nearly 50% PNG children affected.
 - e. 62% of children under 6 months are exclusively breastfed
 - f. There is correlation between outreach service & immunisation coverage. As outreach activity declined,

so too, immunisation coverage

- II. Mother to child transmission rates of HIV at the PMGH clinic are 25%, with 17% in Mt Hagen: should be <10%, and most countries achieve 5%.
 - a. HIV is a cause of 8% of post-neonatal children deaths in hospital. It is estimated that more than 2% of neonates are born with positive syphilis serology.
 - b. Prevention of Mother to child transmission requires testing in pregnancy to determine presence of virus & providing treatment during pregnancy to minimise risk of vertical transmission (to child).
- III. Adolescent women:
 - a. Adolescent women 12% begun childbearing; rural teenagers (13%) start childbearing earlier than urban teenagers (10%); Adolescent pregnancy is associated with two to five times higher maternal mortality, higher neonatal & infant mortality; married teenagers, 18% using some type of family planning, injectable (6%) and implants (9%).
- IV. Family Planning:
 - a. 37% of women currently use some form of family planning. There remains an unmet need of a further 26%.
- V. ANC:
 - a. Of those women who died after delivery that had delivered in community, 57% received no ANC, compared 17% of those who delivered in a facility; 76% women who gave birth in last 5yrs attended at least one antenatal visit: half (51%) of rural women and 85% of urban women deliver in a health facility

2020 SPAR – Children

- *Pneumonia in Children <5: CFR (%) - % children <5 admitted to a health facility with pneumonia & die during that admission; Good quality care – oxygen, early & effective use of antibiotic minimise deaths. (Best performing Western Highlands & Hela; Poor performing – Oro)*
- *Malnutrition severe: % children <5 attend MCH Clinic are moderately (60-80% weight for age) or severely (<60% weight for age) malnourished. Average: 15.9%*
- *Low birth Weight: % of live births in health facility weight less than 2500grams: (Best performing AROB, Milne Bay; Poor performing – Gulf & East Sepik)*
- *Diarrhoea (<5/1,000): number of children under 5yrs who seek care for diarrhoeal illnesses as a proportion of every 1000 children under 5yrs. Diarrhoeal illness serves as indicator of water quality, food hygiene, personal hygiene. (Diarrhoeal disease in children increasing nationally – Highlands, Western, Gulf & New Ireland)*
- *Immunisation: essential component for reducing child mortality; coverage used to monitor coverage & quality of child health services.*
 - *Measles (46% coverage) leading cause of childhood mortality from vaccine preventable diseases; indicator provides good measure of health system performance; (Best performing NCD & Milne Bay; Poor performing – Western & Oro)*
 - *Pentavalent vaccinations coverage (48% coverage): proportion of children under 1yr received 3 doses of DTP-Hib-Hep B (Pentavalent) vaccine; Best performing NCD & Milne Bay; Poor performing Western & Gulf).*
- *Outreach Clinics/1000 children: key platform for preventive child health programs, opportunity for individual, community, health education; increased effort needed to strengthen outreach in rural areas & reverse decreasing trends especially in ESP, Morobe, Gulf & Western.*

2020 SPAR Maternal Health

- *ANC (1st visit%): indicator of access use of health care during pregnancy; % of pregnant women attended at least 1 ANC visit at hospital, health centre or outreach clinic;*
- *Supervised deliveries: proportion of births at health facility attended by skilled HW, measuring maternal mortality (sisterhood method); supervision of delivery is used as a proxy. Low supervised deliveries are a huge concern & must be addressed;*
- *Family planning – Couple yrs. Protection (CYP/1,000, women reproductive age): modern contraceptive methods used for every 1,000 women of reproductive age; more advocacy & effort needed to raise prominence of family planning as a major health & development issue. (Best performing Milne Bay & ENB, Poor performing – Sth H'lands, Enga)*
- *Referral: rates per 1000 births.*

INFECTIOUS – COMMUNICABLE DISEASES

Sometimes, a greater than expected number of cases of a disease occurs in a group of people living in a wider community or within population groups. This is called a disease cluster or sometimes a cluster of communicable diseases.

Communicable diseases, which are diseases that can be spread from one person to another, often occur in clusters. Examples include TB, Malaria, whooping cough, chlamydia and HIV. In PNG 9% of hospital admissions relate to Tuberculosis and include children & multidrug resistant TB.

Addressing communicable diseases as a 'cluster' offer opportunities and challenges to effectively address high-impact communicable disease through:

- effective, people-centred, integrated quality services across the healthcare continuum
- strategic information to improve the response;
- multisectoral collaboration and actions, and innovation to accelerate response.

The significant infectious-communicable diseases for people living in PNG includes:

- HIV & STIs
- Malaria (accounts for 4% of total admissions)
- Adult Febrile Illnesses inc: Meningitis (accounts for 3% of total admission)
- Diarrhoeal symptoms– acute & chronic (11% of total admissions)
- Neglected Tropical Diseases
- Pandemics and Emergency Preparedness.

Supporting health information:

Global Indicators:

End Pandemic of Aids, TB, Malaria, Neglected Tropical Diseases and Combat Hepatitis, Waterborne Diseases & other Communicable Diseases Malaria

PNG Health information:

Infectious diseases remain a major cause of death (21.3%) with increased malaria, especially in children under five (8.8%). HIV rates are stable at 0.83% however there is increasing drug resistance at 18.4 % which is one of the highest in the world. There are 27,000 new cases of TB each year, with 3.4% of these new cases with multiple drug resistance. TB treatment completion rates at 69% remain below the needed threshold for control

PNG eNHIS information:

Morbidity: leading reasons 2018 malaria, skin disorders and diarrhoeal disease, very little change over the previous 5 years.

- I. TB: concentration of TB several provinces, NCD, Western, Gulf, West New Britain; a case notification rate above 600/100,000 - highest in Western Pacific Region.
- II. HIV: Women (15 yrs. & older) account 59% of infected persons & 7% being children less than 14 yrs. Prevalence 14.9% female sex workers (FSW) & 8.5% among men-who-have-sex-with-men (MSM) & transgender (TG) people in NCD. The burden is highest in Highlands region. Focus: that 90% of people living with HIV know their status, 90% of those who test positive are started on Anti-retroviral Therapy (ART) with 90% of those on treatment achieving viral suppression.
- III. Malaria: prevalence 7.1% of people living below 1600m altitude: 8.8% of children (less than 5 years) are infected.
- IV. Neglected Tropical Diseases (NTD): high burden upon community; Lymphatic Filariasis, Dengue, Treponematosi, YAWS, congenital syphilis, Trachoma and Buruli ulcers; Dengue seroprevalence was very high with an overall rate of 85.3% which increases with age. Yaws entrenched as public health problem in Islands region, Madang & Morobe provinces, increasing trend over past 7 years.

2020 SPAR:

Malaria: rate of confirmed cases (confirmed by slide or RDT) & probable (unconfirmed) cases (cases that were not tested but treated as malaria) per 1,000 people; (Poor performing – East Sepik)

NON-COMMUNICABLE DISEASES

A non-communicable disease is a disease that isn't infectious, which means it can't be passed from one person to another. Examples include cancer, cardiovascular disease, asthma and diabetes.

Reports of possible non-communicable disease clusters cause concern in the community, such as:

- Respiratory, & Related Disorders
- Endocrine Disorders (diabetic)
- Cardiovascular (Ischemic heart disease & VCA)
- Cancer (lung, liver, cervical)
- Mental, Neurological & Substance Use Disorders
- Musculoskeletal Disorders
- Oral Health
- Congenital & Genetic Disorders.

Supporting health information:

Global Indicators:

- ▲ Reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being
- ▲ Strengthen the prevention and treatment of substance abuse, including narcotic drug abuse and harmful use of alcohol

PNG health information:

Morbidity: leading reasons 2018 upper and lower respiratory tract concerns, very little change over the previous 5 years.

Cancer: highest estimated burdens of cervical cancer globally; followed by cancer of the oral cavity (11%), breast cancer (8%) and liver cancer (7%). These 4 cancers types account for 44% of all cancer discharges.

PNG: Non-communicable diseases (NCD) account for most deaths (40.3%) with complications from diabetes, ischaemic heart disease, lung cancer and strokes the major contributors.

Global Indicators	PNG	Trauma - Injury
▲ Halve the number of global	Injury: accounted for 19.1% of deaths	i. Trauma & Emergency

deaths and injuries and from road including financial risk protection ▲ substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	(small study); Violence: 59% women aged 15 – 49 experienced physical violence or sexual violence, a quarter of all women have experienced both physical and sexual violence. 19.1% of all deaths (male – 24%, female 13%) were a result of injury; Nearly half result for violence, falls, homicide, a sixth from motor vehicle accidents.	Management (discharges) ii. Injury Prevention, iii. Gender based violence (strategy) iv. Environment Improvement
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SURGICAL SERVICES AND ANAESTHESIA CARE

There is still a lack of access to safe, affordable and timely surgical and anaesthesia care at rural levels of care across PNG. This leads to preventable mortality, unnecessary disability and uncorrected deformity that negatively impact health and economic activity.

There is a need to strengthen Emergency and Surgical Care and Anaesthesia as a part of Universal Health Coverage. In order to achieve this and improve the ability to deliver emergency and surgical care, it is necessary to measure access in terms of capability, capacity, timeliness, safety and affordability

Technological advancement is rapidly changing the field of surgical and procedural interventions. The past decades have seen significant changes, particularly with the growth of robotic surgery, increasing use of imaging in surgery and the development of minimally invasive techniques and interventions. There has also been a broader application of laparoscopic and endoscopic techniques.

The provision of surgical services is dependent on workforce support for anaesthetic care, diagnostic services, etc. Considerations is given to complexity, cost and ethical implications, critical for surgical and procedural services.

Recovery after surgery requires protocols to ensure surgical efficiency and enhanced patient outcomes by reducing length of stays and increasing day only and extended day only procedures.

Surgical and procedural services are provided need to be planned, prioritised to meet projected surgical demand to further develop surgical and procedural services.

Supporting health information:

Surgery: diseases and conditions requiring or amenable to operative or manual procedures – across spectrum of care

PNG: Rise of diabetes & CVD (leading to stroke) will contribute to *increased disability including visual impairment & amputation*;
Visual impairment over 50 yrs. estimates 40,746 blind (5.6%), further 21,519 severely visually impaired (2.9%), 79,463 moderately visually impaired (10.9%).

- i. Surgery,
- ii. Dental – Oral Health
- iii. Rehabilitation, Allied Health, Orthotics
Prosthetics
- iv. Palliative Care & Pain Control
(cross reference with Cancer management)

Clinical Support Services

- i. Pathology
- ii. Blood Transfusion Services
- iii. Radiology
- iv. Pharmacy- Medical Supply Chain Management (Drugs, consumables, cold chain)

HEALTH SYSTEM STRENGTHENING

Health systems strengthening work emphasizes a systematic, evidence-based approach designed to bring about significant improvement to improve:

- Patient and population health outcomes, particularly among the poor
- Efficiency and effectiveness of systems and processes of care
- Social responsiveness and accountability.

Supporting health information:

Global	SPAR	PNG
▲ Health service coverage & financial protection coverage. ▲ Achieve UHC: financial risk protection, access to quality health-care services, access to safe, effective, quality & affordable essential medicines & vaccines for all. ▲ Use promotive, preventive, curative, rehabilitative and palliative health services they need & of sufficient quality effectively.	Access: OP Utilisation: ▲ Outpatient visits/persons/year : Outpatient per person; Average: Medical supplies: ▲ Month with stock - % stock out: Average:	<u>Funding</u>: per capita Health Expenditure (GoPNG) expenditure has declined from K123/person (in 2011) to K67/person (in 2018) <u>Workforce</u>: population ratios reduced across all disciplines from 1.27/1,000 population in 2011 to 0.97/1,000 in 2018. Staff ceilings not being met, with overall 27% vacancy rate. Pre-service training institutions are well short of capacity to produce required workforce. PNG lowest ratio of both doctors & nurses per 1,000 population across the Pacific with 0.07 doctors & 0.53 nurses per 1,000 population. <u>Medical supply</u>: continues punctuated by interruptions stock levels of essential medications; present first line treatment drugs for malaria, artemether-lumefantrine, available in 59% of Level 3 & 4 public health facilities. Similarly, with vaccines, availability index for Level 3 & 4 facilities was 73.7. <u>Facilities</u>: 30 – 60 % Level 3 and 4 facilities are in need of significant remediation. Services to the most remote rural populations have decreased (48% aid post closures), although investment in development of new rural facilities (32 Community Health Posts). <u>Service utilisation</u>: Utilisation of facilities is declining including Outpatient attendance, Antenatal care, supervised delivery, outreach and Immunisation coverage.

In Summary

The 'Levels of Care and Health Services'	Health System Strengthening
Community Initiative Community: Village Health Assistants Roles, Competencies Public Health Interventions Pre hospital Services: Ambulance – Retrieval Patient Referral Clinical Care Radiology & Ultrasound Pathology & Blood Transfusion Rehabilitation: Allied Health, Orthotics, Prosthetics Networked Services & Societies and	Leadership & Management Governance • Critical Risks • Compliance IPC & Waste Management Health Workforce + Administrative, Ancillary • Workforce Productivity • Licencing, Credentialing within National hospital, and PHA • Education, Inservice – Continuous Professional Development Medical Supplies Distribution, inc: Essential D&S, ANC Syphilis / HIV Test, Reagents Equipment, and Biomedical Engineering, Infrastructure, Facility Management, Minimum Standards Innovative & Evolving ICT Solutions and Health Information

DEVELOPMENT OF HEALTHCARE SERVICES RESPONSIVE TO NEED AND AVAILABLE TO ALL

Primary Healthcare Services

Services provided by primary health care vary, from prevention and health promotion activities, screening, to the treatment and management of illness. Primary health care is often the 'best setting for the prevention and management of chronic and complex health conditions', delivered in a variety of settings (primary health care providers, health centres & hospitals, and the broader community) through communication technology, delivered by various health professionals.

Urban and Rural Healthcare Services and Level 2-4 Health Facilities: Rural Hospital and Health Services:

These services are responsible for the provision of PHC and clinical care in a defined geographical area and ensure established referral pathways to ensure patients receive the right care in the right place at the right time.

Secondary Healthcare Services – Level 5 Hospitals & Level 6 National Referral Specialist Tertiary and Teaching Hospital

PHC does not operate in isolation and is part of a larger system involving other services and sectors and so can be considered the 'gateway' to the wider health system. Individuals can be directed from one primary care service to another, and from primary services into secondary and other health services, and back again.

'Secondary Care' is medical - clinical care provided by a specialist or facility upon referral by a primary healthcare provider. Hospital emergency departments are a critical component of hospitals and provide care for patients who have an urgent need for medical or surgical care.

HOW TO DETERMINE THE SERVICE LEVEL

A health facility or health service capacity to deliver a particular level of health service depends on the presence of medical, nursing, allied health and ancillary health care personnel supported by public health officers who have the qualifications, skills, experience and competence, compatible with the defined level of care required in the defined service cluster or group and has also the appropriate level of support services.

It also depends on the availability of medical supplies and equipment e.g. drugs, medicines, ventilators, CT scanners, and non-medical requirements, e.g. consumable supplies.

The specific factors determining service capability level differs according to the service group but generally is a combination of:

- Service complexity;
- Catchment community and Patient characteristics; and

- Support services availability and health workforce capability.

SUMMARY:

Delineating '*Levels of Care and Defining Health Services*' is a planning matrix that assists in the development of the National Referral Specialist Tertiary and Teaching Hospital and PHA 5year strategic health services development plan.

This is achieved by comparing existing and required services (public health, primary health and clinical care and support services); identifying necessary changes to meet requirements and reflecting on either programs or projects.

All PHAs, National Referral Specialist Tertiary and Teaching Hospital, health partners and other health services providers are to have in place quality assurance/continuous quality improvement programs.

The National Referral Specialist Tertiary and Teaching Hospital and the PHA model of care, delineated levels of care and defined health services link to and support the 'National Quality Standards for Health Services in PNG' and the 'Papua New Guinea Health Service Accreditation Program'.

All PHAs and the National Referral Specialist Tertiary and Teaching Hospital are to ensure implementation of all Levels of Care, and to support their health partners participate in district and provincial quality management and monitoring processes, working together towards meeting the National Quality standards.

All PHAs, National Referral Specialist Tertiary and Teaching Hospital, health partners and other Health services providers are to utilise current best practice clinical techniques and public health interventions, and ensure the effectiveness of the services delivered are monitored and evaluated against the following outcome indicators which Papua New Guinea has in place:

- National Referral Specialist Tertiary and Teaching Hospital, PHA provincial and local quality assurance mechanisms including peer review mechanisms;
- External National Referral Specialist Tertiary and Teaching Hospital, PHA review including health service accreditation set against national quality standards;
- Continuous improvement approach to health services delivery, practice change and support services;
- Continual health and support staff training, upskilling and professional development; and
- An emphasis on obtaining and using data/information/feedback from public health programs and patient care to improve the safety and quality of care of health services for the people of Papua New Guinea.

HEALTH SYSTEM STRENGTHENING

Health systems strengthening work emphasizes a systematic, evidence-based approach designed to bring about significant improvement and strengthen the performance and interconnectedness of the PNG healthcare system building blocks identified in the NHP 2011-2030:

- Organizational Governance and Leadership,
- Health Workforce,
- Health Information,
- Health Infrastructure, Equipment And
- Commodities (Essential Medical Supplies), And
- Health Financing

ORGANISATIONAL GOVERNANCE AND LEADERSHIP

The development of a '*PHA Operations Manual*' is a guide for ensuring all **Boards, Chief Executive Officers and Executive Management teams commit to ensuring strong corporate governance and leadership is applied** at all levels of the PNG health system.

National Referral Specialist Tertiary and Teaching Hospital and PHA Boards will do this through their devolved powers and accountabilities to its Chief Executive Officer and Executive Directors, by seeking assurances that make the PHA accountable to the local community within the province for:

- Coordinating and monitoring the implementation of the National Health Plan (NHP), the National Health Service Standards (NHSS), and the National Referral Specialist Tertiary and Teaching Hospital and PHA's 5year Strategic Health Services Development Plan, with relevant supporting plans;
- Providing advice to NDoH on the provision of adequate resources to carry out the implementation of the NHP, NHSS, and the PHA 5year Strategic Health Services Development Plan;
- Administering and expending moneys appropriated by the PNG State and development (Donor) partners;
- Accounting for the funds appropriated to it by the National Parliament for the delivery of the National Referral Specialist Tertiary and Teaching Hospital and PHA health services, in the manner set out in the Public Financial (Management) Act and by the formulation of annual financial statements and reports to the Minister for Health;
- Effectively administering and maintaining the National Referral Specialist Tertiary and Teaching Hospital and PHA and all of its facilities, equipment and assets;
- Providing or assisting with education, instruction or practical training and ongoing development of its professional staff and other employees;
- Providing facilities for teaching, instruction, research or post-graduate studies for health workers;
- Encouraging international standard research and experimentation into any areas of public health services, medical-clinical activities or paramedical activities;
- Consulting and cooperating with appropriate health partners, other authorities and organizations, associations and persons on matters related to National Referral Specialist Tertiary and Teaching Hospital and PHA health service and support activities;
- Engaging with and assisting local authorities in the provision of community health education;
- Providing relief to sick and injured persons through the provision of patient centred safe quality health care and treatment;
- Promoting, protecting and maintaining the health of all communities-villages within the province;
- Encouraging local community participation in planning and in the decision-making processes of the PHA and the National Referral Specialist Tertiary and Teaching Hospital;
- Providing information, awareness, knowledge and programs in the field of public health;
- Delivering public health services appropriate and acceptable to the local community;
- Delivering curative-clinical services from the premises of the National Referral Specialist Tertiary and Teaching Hospital, a PHA provincial and district hospitals, rural – urban health facilities or other place as the case may be, which are appropriate and acceptable to the local community;

- Advising the Provincial Government on policy matters relating to health, in particular administration of any legislation relating to health matters;
- Maintaining effective liaison with NDoH, all provincial districts, and with the private sector;
- Conducting inquiries into the operation of all health facilities within the province and their provision of health services and programs, and make recommendations for improvement; and
- Carrying-out any other functions that are delegated to it by the Minister for Health, the Secretary of Health, or by the Provincial Government.

The development of a new '*Monitoring Department Manual*' for the revised functional role of the National Department of Health (NDoH) in terms of monitoring and evaluating the whole of the PNG health system, seeks to ensure that all threads of health governance, stewardship, health policy, planning, performance and ongoing development, are aligned and integrated to make certain the delivery of safe quality health services across the whole of PNG.

CLINICAL PUBLIC HEALTH GOVERNANCE FRAMEWORK AND MANUAL

The development of the '*Clinical and Public Health Manual*' for Provincial Health Authorities (PHAs) and the National Referral Specialist Tertiary and Teaching Hospital in Papua New Guinea is a guide for ensuring all Boards can commit to the delivery of safe quality health services delivery within their province.

PHA Boards will do this through ensuring strong clinical and corporate governance, committed partnerships and effective oversight supported by this Clinical and Public Health Governance Framework, and implementation of associated systems, standards, protocols and processes.

The National Referral Specialist Tertiary and Teaching Hospital and PHAs are responsible for the quality and safety of health services provided by all health facilities, at all levels, by all clinicians, healthcare providers, health partners and non-clinical staff as reflected in these NHSS.

The Clinical Governance Framework is required to be embedded into PHA organisational structures and functions at the National Hospital, Provincial Hospitals, District Hospitals, Urban and Rural Health Services, and within Public Health Programs through the leadership of the PHA Board, Chief Executive Officer and Senior Executive Management team taking responsibility and accountability for integration of Health Services and Public Health Program delivery throughout the province.

Clinical and Public Health Governance is the set of relationships and responsibilities established by NDoH, each individual PHA, relevant Executive Management teams, clinicians, healthcare providers, patients-consumers and other stakeholders, to ensure good clinical and health outcomes.

It ensures that the community and healthcare providers can be confident that healthcare systems, standards, processes and procedures are all in place and being followed, to deliver safe and high-quality health care, programs and support services which are continuously improving.

It also requires the National Referral Specialist Tertiary and Teaching Hospital, all PHA Chief Executive Officers (CEOs) and Senior Executive Management teams take both individual and

collective responsibility and accountability for the cost-effectiveness of Public Health Programs, Primary Health Care and Clinical Health Services and report on a regular quarterly basis directly to the Board.

It is important that CEOs take prompt leadership and management action on significant issues and concerns to ensure a work-culture of continuous quality improvement in all clinical, public health programs and primary health care services, in all health facility levels, including that within health partner organisations.

Clinical Governance is an integrated component with corporate governance and ensures that everyone – from frontline clinicians to managers and members of Boards – are all accountable to patients and the community for assuring the delivery of health services that are safe, effective, integrated, of high quality and continuously improving.

All clinicians-public health-healthcare providers are held accountable for conducting evidenced-based practice, evidenced-based public health interventions and that decisions are made on the best available data/information (evidence) and resources.

The healthcare delivery goal is to ensure all patients (the 'right patient') and the broader community, receive the 'right care', in the 'right place', at the 'right time', by the right staff, and in the 'right way'.

The National Referral Specialist Tertiary and Teaching Hospital and PHAs are to ensure all health workers are continually being skilled to deliver quality primary and secondary healthcare and public health programs, to ensure the allocation of adequate support resources, to provide health services more equitably, and to inform their patients/consumers and communities about the services they can reasonably expect to access and receive within the province.

PATIENT REFERRAL GUIDELINE

Formalising Patient Referral Pathways

The development of a National Patient Referral Guideline for Health Services in Papua New Guinea is a guide towards formalising a well-structured and efficient patient referral system that supports the implementation of the NHSS.

This guideline applies to the National Referral Specialist Tertiary and Teaching Hospital, all PHAs, health partners, other health service providers, and all health staff who provide patient care and initiate approved patient referrals, as well as hospitals that receive patients from rural and urban health facilities, and those health facilities that discharge patients back to rural or urban health facilities, urban clinics or to their home.

The guideline provides a clear direction to effectively and efficiently manage the patient referral system within Papua New Guinea and is in line with the National Health Service Standards 2nd edition 2021-2030.

"The process of referring patients from primary health care levels to the next appropriate level for

continuous provision of health care has been identified as an integral component of an effective health care delivery system”.

The guideline addresses the following:

- role of the initiating health facilities as to *‘who gets referred’*
- dialogue between healthcare providers to *‘confirm appropriate referrals’*
- encourages healthcare workers to *‘treat and stabilize patients adequately’* prior to referral and to *‘consider the decision of when to refer’*.

Furthermore, the guideline addresses the information requirements for the patient and their family, where all health workers are encouraged to show empathy, and the need to consider the cost of transport, treatment and family accommodation, ensuring both the patient and family understand the reasons and importance of their referral and the risks of non-referral.

All PNG healthcare workers are held accountable to refer patient’s safely and appropriately by following these guidelines of care.

A good referral system can help to ensure patients receive optimal care at the appropriate level, hospital care is cost-effective, patients who most need specialist health services can access them in a timely way, and primary health services are well utilised, and their reputation is enhanced.

HEALTH WORKFORCE MINIMUM STANDARD FOR HEALTH SERVICES

Prioritising Workforce Capacity, Training, Ongoing Professional Development and Performance

National Referral Specialist Tertiary and Teaching Hospital and PHA health workforce minimum standards core component include that:

- there are sufficient staff to provide services, where and when they are required.
- staff are appropriately qualified, sufficiently experienced and competent to provide health services and
- staff maintain their ongoing competence and qualifications to provide those services.

Competence relates to skills, knowledge and attitude of the individual. An appropriate credentialing process is in place in the National Referral Specialist Tertiary and Teaching Hospital and all PHAs to ensure staff with the appropriate skills are working to an agreed scope of practice. A health workforce supports a patient-focused, safe and quality conscious work environment, integral to providing competent, evidence-based, health care services by a PHA.

The demand for increasing health workforce capacity has arisen due to factors such as the current '*population growth rate*' and the subsequent increasing need for health care services, and which has also been compounded by workforce shortages, the ageing of the existing workforce, and inability to adequately 'upskill' and develop all health workers on a continuing basis.

All PHAs are now required to establish an effective '*Health Training and Resource Centre*' (HTRC) designed to develop and deliver mandatory and continuing training, upskilling, and ongoing professional and technical development for all health workers and corporate support staff.

These PHA HTRC centres should be adequately staffed with well-qualified '*human resource development officers*' (and not just simply by former in-service nursing training officers), who are to be responsible and accountable for:

- the conduct of regular annual training needs analysis (TNAs) for all PHA staff (clinical and non-clinical), in line with annual performance appraisals and subsequent training requirements.
- the facilitation of a PHA training plan and implementation of an annual calendar of training, skilling and capacity-building events, in line with the requirements of clinical and non-clinical managers.
- The oversight of all ongoing professional and technical development activities within the PHA, including short-term transfers and/or secondments for capacity-building (eg: rural nurses and CHWs gaining practical operational experience within hospital facilities, and hospital nurses and CHWs gaining operational experience in a rural setting, etc.)
- The promotion and application of training scholarships.
- Oversight use of the ICT resources/facilities of the centre for research and ongoing continuous quality and service improvements.
- The day-to-day operations of the PHAs fully functioning training and resource centre, including development of planning, budgeting, monitoring, evaluating and reporting to the PHA Board, through the CEO, on all matters related to training, upskilling, and ongoing professional and

technical development for all health workers and corporate support staff.

MINIMUM STAFFING STANDARDS FOR HEALTH FACILITIES LEVEL 1 TO 4

Building on from the work of the Workload Indicators of Staffing need (WISN) methodology in PNG and the NHSS Staffing Model for rural health services.

Table 1 below shows the minimum staffing standards for health facilities Level 1 to 4.

This minimum staffing standard is utilized in the projected staffing workforce targets for rural health service in the National HRH Strategic Plan 2021 – 2030.

Minimum Staffing Standards as per NHSS for NHP 2021 - 2030						
Level of Health Facilities	Category of staff	Minimum standards	Justification for Additional Health Workforce for Rural Health Services	Total staff	WISN	Catchment Population
Census Unit / Wards	Village Health Assistant (VHA)	3		3	=3 or >3	1000
Level 1: Health Post	Community Health Worker (CHW)	1	1 for VHA community program activities if implemented	2	= 2 or > 2	2000
Level 2: Community Health Post	Nursing Officer	1	Family health Services, Public Health, Ambulatory Care & 24hr inpatient care including supervised deliveries requires total 5.5 FTE	5.5	>1	6000
	CHW (at least 1 trained with EMOC skills) (EMOC - Emergency Obstetric Care)	4			= 5 or > 5	
Level 3: Health Centre	Health Extension Officer (HEO) - 1 OIC	1	Based on catchment population, demand for services; meeting ambulatory care, supervised deliveries and inpatient targets; may have x ray unit - Medical Imaging Technician; may have dental clinic - Dental Therapist. L3 Support team includes Clerk for medical records management, Hygiene Workers, Transport Drivers, Security & Handyman.	16	= 2 or > 2	15000
	Nursing Officer	4			= 4 or > 4	
	Midwife	2			>1	
	CHW	7				
	Lab Technician	1			>1	
	Pharmacy Assistant	1			>1	
Level 4: District Hospital	Rural Health Medical Specialist (SMO) - OIC	1	As above + new L4 Hospital: with Rehabilitation unit - Physio Assistant; with Biomedical Workshop - Biomedical Officer; with Medical Records Management system - Health Information Officer; Support Team: Site Facility Manager, Administrative Officers - Clerks (3); Kitchen/Laundry Team (3); Hygiene team (5) determined by sq meter areas of infrastructure; Maintenance Team (2); Drivers, security and grounds management team (5); District Public Health team: at a minimum Public Health Family Health Officer (Men, Women, Maternal Newborn, Family Planning, Nutrition, Child & Adolescent Health), Environmental Health Officer/s, Disease Control and Surveillance Officer (Communicable & Non Communicable Disease Inc Mental Health), Health Promotion / Prevention / Community Development Officer support for VHA programs. : Target activity 50,000 annual / 160 occasions of service daily - 700 inpatient admissions annually and 500 supervised deliveries annually.	47	>1	40000
	Medical Officer	1			>1	
	HEO	2			= 2 or > 2	
	Senior Nursing Officer SIC inc QA - IPC Govern	1			>1	
	Nurse Unit Managers inc Evening Supervision	3			= 3 or > 3	
	Nursing Officer (specialist nurses)	5			= 5 or > 5	
	Nursing Officer (general)	10			= 10 or > 10	
	Midwife	3			= 3 or > 3	
	CHW	10			= 10 or > 10	
	Porter	1			>1	
	Pharmacist	1			>1	
	Pharmacy Assistant	1			>1	
	Supply - Logistic Officer	1			>1	
	Lab Scientist	1			>1	
	Medical Lab Technician	1			>1	
	Blood Bank Nursing Officer	1			>1	
	Dental Officer	1			>1	
	Dental Therapist	1			>1	
	Anaesthetic Scientific Officer	1			>1	
	Radiographer	1			>1	

STAFFING STANDARDS FOR NHSS HEALTH FACILITIES LEVEL 5 AND 6

It is acknowledged that the complexity of health care increases with the health facility level especially from NHSS level 4 to 6. In this context, the NHSS Clinical Toolkit will specifically report by NHSS level, the specialist and subspecialist hospital clinical data not reported by eNHIS.

Aligned to the National HRH Strategic Plan 2021 – 2030, the data from NHSS Clinical Toolkit, Human Resource Information System (HRIS) and eNHIS will provide data for the Health Workforce Standards & Monitoring System (HWSMS) to generate Workforce Assessment reports for individual health facilities under PHAs. This report will show whether the current staffing available is able to cope with the health care demand and what needs to be done.

PRIORITISING WORKFORCE CAPACITY, TRAINING, ONGOING PROFESSIONAL DEVELOPMENT AND PERFORMANCE

Ensuring the availability of competent, sufficient and qualified health workers in the right place is essential for delivering quality health services to the population.

A health workforce that is supported in an enabling work environment is integral at the National Referral Specialist Tertiary and Teaching Hospital and PHAs and health facilities.

The demand for increasing health workforce capacity has arisen due to factors such as the increasing population growth rates in the country, an ageing population and increased demand for health care services.

This has also been compounded by workforce shortages, an ageing workforce in service and inability to adequately ‘upskill’ and develop all health workers on a continuing basis.

NDoH, National Referral Specialist Tertiary and Teaching Hospital, PHAs, health facilities and the collaborating partners are now required to:

- **Effectively plan for health workers using evidence-based methodologies that are scientific and use workloads to provide evidence that goes beyond the use of population and density of health workers numbers. The Health Workforce Standards and Monitoring System (HWSMS) will be the tool used to guide the National Referral Specialist Tertiary and Teaching Hospital and PHAs in optimising workforce utilisation and workforce planning**
- **Improve health workforce information by having a functioning HRIS/database that is interoperable with other systems like the NHIS to generate evidence for decision making**
- **Scale up education and training to produce adequate and competent health workers as well as strengthening the regulatory capacity**
- **To establish an effective ‘*training and resource centre*’ designed to develop and deliver continuing training, upskilling, and ongoing professional and technical development for all health workers and corporate support staff. These centres should be adequately staffed with qualified ‘*human resource development officers*’ (and not just simply by former in-service nursing training officers), who are to be responsible and accountable for:**

- **To conduct of annual training needs analysis (TNAs) for all PHA staff (clinical and non-clinical) in line with annual performance appraisals and subsequent training requirements. This should be supported with annual training plans for training, capacity building and development programmes which should be implemented. Provisions for ongoing training, scholarships and incentives like promotions to be instituted**
- **Innovative use of ICT should be embraced for improved quality services**
- **Strengthening the leadership and stewardship through dialogue and partnerships to ensure functional PHAs that can plan, budget, monitor, evaluate and report all activities of health workers**

INCREASED PRODUCTIVITY

PNG needs to accelerate the attainment of UHC and health SDGs by ensuring equity in access, quality and efficiency of health services through all of its health workers.

Health worker shortages, inadequate capacity for ongoing training and education, maldistribution across facilities and provinces, inadequate skills mix, weak leadership, management and stewardship have all negatively impacted over time on the productivity of PNG health staff to deliver health care successfully, which in turn, has ultimately affected the ability of the Country to meet its national and global health targets.

In order to increase health worker productivity, the National Referral Specialist Tertiary and Teaching Hospital and PHAs should ensure that all obstacles should be prevented through improving training and capacity-building, improving the effective allocation of adequate workloads and skills-mix, improving individual work environments, improving performance management, improving communication and feedback channels between NDoH, National Referral Specialist Tertiary and Teaching Hospital, PHAs and all health facilities, and finally improving the provision of the right tools and equipment for health workers to use.

Most times workforce productivity problems contributing to health service delivery gaps can be improved with no, or few, additional resources or external support even at health facilities levels.

Addressing these health workforce barriers through improving health worker performance and productivity is an important strategy toward achieving and sustaining the new NHP goals, KRAs, and the NHSS.

National Referral Specialist Tertiary and Teaching Hospital and all PHAs in collaboration with their respective health partners, are now required to:

- **identify and clearly understand the underlying causes of possible health workforce productivity and performance problems;**
- **develop remedial intervention strategies for employee and organizational development, to monitor and evaluate those strategies; and**

- **report quarterly to the Board, through the CEO, on achievements at improving health worker productivity and performance at all levels of the organization.**

NDoH is required to support the National Referral Specialist Tertiary and Teaching Hospital, all PHAs and their collaborating health partners to ensure that there is adequate supply of health graduates (medical specialists, urban and rural doctors, nursing, CHWs, corporate/technical support, etc) qualifying annually from all PNG Universities, who are able to be recruited and deployed appropriately.

These Graduates should also be oriented to the work stations and duties, mentored and scheduled for health sector-wide employee and organization training and development programs related to supporting ongoing capacity and capability building activities.

WORKLOAD DISTRIBUTION

Workload distribution is the process of efficiently distributing and managing work across the health worker cadres based on their skills, knowledge and capabilities.

Increasing demand for high-quality health services has led to an increased pressure on human resources planners to allocate the optimal number of health workers in different service areas and different geographic areas, and simultaneously, promote the desirable population coverage.

Human resource allocation in wrong numbers or wrong proportions or inappropriate geographic areas can lead to a bad effect on the quality-of-service provision, an increased dissatisfaction among clients and even staff. Ultimately, it can be an obstacle to the achievement of the health goals such as UHC and SDGs. The essential need is to estimate the optimal number of employees to ensure effective and efficient use of limited resources.

When workload management is successfully done through describing workload components appropriate for each health worker, performance is maximized and qualities of health services are improved significantly. Currently, a manual with workload components per cadre is under development for all levels of care by the Workforce Standards team of NDoH.

ALIGNMENT OF WORKFORCE TO WORKLOAD

Workforce alignment is a valuable practice in human resources for health management.

High-performance work system fosters high performance through positive effect by aligning the health workers to their key competencies to perform the tasks and roles. Health service delivery improves if workforce with the right skills and behaviour are deployed in the right places doing the right tasks.

NDOH, National Referral Specialist Tertiary and Teaching Hospital and PHAs will strive to ensure that the right health workers are deployed in the right places with the right skills and attitudes performing the right tasks at the right time and in the right environment.

National Referral Specialist Tertiary and Teaching Hospital and all PHAs in collaboration with their respective health partners, are now required to:

- ensure correct optimal allocation and distribution of health workers based upon evidence-based workload indicators;
- ensure the correct “skills-mix” and workforce alignment to workload, within all health facilities, based on the NHSS “*Workforce Indicator Staffing Needs*” (WISN) model; and
- report quarterly to the Board, through the CEO, on achievements at improving health workforce to workload requirements, including transfers of health workers to meet workload levels of the organization.

ESSENTIAL MEDICAL SUPPLIES, PROCUREMENT AND SUPPLY SYSTEMS

Improving Essential Medical Supplies Procurement and Supply System

The status of the medical supply chain in PNG needs considerable improvement.

There are constant shortages, over stocking of some items, and large-scale wastage occurring on account of product expiry prior to use.

Inaccurate quantification of supply requirements has been identified as the root cause of the imbalance in stock holdings while short comings in other areas such as procurement, warehousing and stock management, and distribution have contributed to the weaknesses in the supply chain.

Non adherence to PNG Standard Treatment Guidelines, misuse of drugs, corrupt activities throughout the medical supply system have impacted on the shortcomings in the system. Staff capacity and capability issues have also compounded the situation.

Although the former National Health Plan 2010-2020 identified some reforms in the overall medical supply chain, it was restricted to reducing the number of Area Medical Stores (AMS) from five down to two AMS.

It is expected, and desired, that the future direction is progressively technology driven and tools such as eNHIS, mSupply and others, are used to support the planning, managing and monitoring of the medical supply chain more efficiently and effectively to provide a better service for the people and the system is managed by staff who are competent, trained and qualified in medical supply chain management.

The NDOH, National Referral Specialist Tertiary and Teaching Hospital, all PHAs and their collaborating Health Partners, are now to ensure drugs and medical supplies are available in the 'right quantity', 'right quality', at the 'right time' and at the 'right price'- in all health facilities consistent with the level of care and services provided by that facility.

All health facilities are to ensure that drugs in particular and medical supplies in general, are:

- **appropriately and adequately stored in conditions that maintain the quality (efficacy) of the products as recommended by the manufacturers of the products**
- **that any 'excess' stocks are transferred to another nearby health facility able to use the stocks, or alternatively, returned to the PHA's provincial hospital pharmacy store; and**
- **that any 'out-of-date' stock are dealt with correctly, by certified destruction (i.e burial or high temperature incineration).**

The new NHP 2021-2030 recognises that the country's medical supply chain needs improvements which have to be more comprehensive and progressive and which also requires a structural re-emphasis whereby the responsibility for forecasting, storage, quantification and distribution management is to rest with all PHAs, as it is considered that they are best placed to do these functions for their respective provinces.

A 10 Year Master Plan for Medical Supply Chain Management in PNG has been designed which identifies key strategies for improving the efficiency and the effectiveness of the PNG medical supply chain, including a supporting implementation plan for the Master plan.

Supporting this Master Plan are guidelines, which have been provided for PHAs to develop their own 5year strategic health procurement plans in order to manage their medical supply chains, and NDoH's Medical Supplies Procurement and Distribution (MSPD) function has been reviewed and revised to reflect the changes within the Master Plan.

The Master Plan's overall strategy involves activities that each individual PHA health facility could do immediately with their present staff and financial resources, and other strategies that are medium to long term which have to be managed both centrally and by PHAs.

Immediate activities for all individual PHA health facilities include:

- cleaning and bringing order to existing medical supply stores within each individual health facility by its own staff, as per Standard Operating Procedures (SOPs) provided for Store and Stock Management; and
- the more medium to longer term activities such as improved forecasting, storage, quantification, procurement, distribution and contract management.

The Master Plan encompasses and emphasises a prioritisation of implementation whereby improving the efficiency and effectiveness of the whole PNG medical supply chain stresses the importance of doing this first within MSPD and in each individual AMS, then in all PHA Provincial Hospitals including National Referral Teaching Hospital, followed by other PHA and Health Partner health facilities.

The simple logic applied that NDoH's MSPD function is responsible for the National Annual Procurement Planning Process and the Procurement process itself, each individual AMS is responsible for storing and distributing to all health facilities including National Referral Teaching Hospital and as PHA Provincial Hospitals account for 60% of the national medical supply budget, this makes it imperative that improving the functioning of these entities would yield substantial positive outcomes.

The Master Plan includes, providing revised supplementary kits to health facilities L1-L4 for an ongoing period of 5 years to supplement what they would be requisitioning from their individual AMS.

In the initial stages, the Master Plan proposes the strengthening of the demand quantification and supply distribution processes within each PHA province.

As a consequence of several enquiries that have been held in regard to shortcomings in the PNG medical supply management system, NDOH and the government of PNG are also currently considering several options for restructuring the whole system, including possible outsourcing

options as a means of improving the future functioning of the whole PNG Medical Supply Chain.

Key components to PNG's Medical Procurement and Supply Chain Management includes:

- Annual Procurement Planning is carried out by MSPD using stock holding, stock duration, usage data sourced from all individual AMSs, and on order, yet to be delivered information, and also quantifications provided for program commodities by the programs themselves;
- Procurement is managed by MSPD with a budget provided by NDOH;
- Warehousing and Supplies are procured under contract by MSPD, stored and distributed to health facilities from the current five (5) individual AMSs;
- Distribution to PHA health facilities is carried out by distributors selected through a formal tender process by MSPD;
- Distribution contractors deliver supplies from suppliers to the individual AMSs and also from the AMSs to individual health facilities throughout the country;
- Supply Management Information System (mSupply) is the system provided to MSPD, all AMSs and most PHA Provincial Hospitals - there are five (5) Provincial Hospitals for installation of mSupply;
- A supply ordering device (a tablet) has been trialled in the Morobe PHA; and
- The medical supply system is managed by competent trained staff; whereas currently managed by personnel who have little understanding of the operation of a supply chain and how each component of it has an impact on the rest of components in the chain.

Irrespective of the PHA health facility level, the following minimum standards are now to be adhered to in all of those facilities:

1. **Storage & Stock Management:** It is critical that all drugs and medical supplies are stored under the right conditions, Stock Management is carried out and record keeping is done to Standard:
 - (a) In respect of stock management and record keeping, where health facilities have been provided with computers and the electronic supply management system, mSupply, stock management parameters such as Minimum and Maximum Stock Levels are maintained in mSupply and re ordering operations are carried out in accordance with these parameters.
 - (b) In PHA health facilities where stock is also held in its Pharmacy, stock records are also maintained; there in a Stock Register to record all receipts and issues from the Pharmacy. In respect of Out Patients & Emergency departments, and in wards, stock holdings are recorded in an Imprest system, where stock replenishments from the Medical Supply Store are carried out and recorded in the individual Imprest.
2. **Stock Recording & Stock Use:** All stock records noted under (a) and (b) above, need to be accurate, clear and comprehensive, and they are all to be treated as accountable documents, that should record the receipts and issues of supplies that are provided to the health facility to serve the members of the public who attend the facility for healthcare.

It is illegal and immoral, to steal drugs and medical supplies provided to health facilities. It is also

injurious to the health of community and individuals if use of drugs stolen from health facilities find their way to local markets and members of the public are enticed to buy and use them. All PHA health service staff must be aware that all medicinal drug use has to be done with qualified clinical guidance.

3. **Stock Ordering:** All stocks of drugs and medical supplies required by individual health facilities are to be based on accurate quantification calculations of requirements covering a minimum three (3) month period, and not merely guesswork.

All individual PHA health facilities are to ensure the submission of their Orders on the times specified by their individual AMS and/or the NDoH's MSPD function.

4. **Stock Supply Systems:** The regular PULL system has been supplemented by a Health Kit "PUSH" system that provides essential drugs and medical supplies to all rural health facilities.

All PHA health facilities are to ensure that any drugs and/or medical supplies provided via a "Health Kit" and which are not used by that health facility, are to be either transferred to another nearby health facility which can use them, or to the PHA's provincial hospital pharmacy store.

5. **Procurement, Warehousing & Distribution of Drugs & Medical Supplies:** The National entity currently responsible for procurement of drugs and medical supplies (i.e. NDOH Medical Supply Procurement & Distribution Branch-MSPD), and the entity responsible for warehousing and distributing to health facilities via contracted distributors (Area Medical Stores -AMS) meet the following MINIMUM STANDARDS for Medical Supply Chain Management:

- **Annual Procurement & Budget Planning:** Ensure that the procurement agency (MSPD) engages in a structured, methodical, approach to annual procurement and budget planning that leads to an Annual Procurement Plan which is reflective of the accurately and comprehensively computed annual demand for drugs and medical supplies in health facilities throughout the country. This plan should be a consolidation of Annual Procurement Plans developed by PHAs.
- **Procurement of Drugs & Medical Supplies:** Ensure the procurement of drugs and medical supplies is consistent with the provisions in the National Procurement Act 2018 of PNG, and is carried out in conformity with fundamental best practices in procurement such as buying at the right price, right quality and within the shortest possible time frames.
- **Monthly Supply Planning:** Ensure a structured and methodical approach to monitoring stock levels and procurement of drugs and medical supplies that may be necessitated due to unexpected shortages, rise in demand and other factors, and to pre-empt, avoid and minimise ad hoc emergency procurement that invariably does not achieve best practice outcomes.
- **Purchase Order Management:** Ensure that purchase order management is carried out efficiently and effectively and that shortcomings in management do not arise and contribute to supply shortages, payment delays to suppliers, and over use of ad-hoc emergency procurements.

- **Timely Payments to Suppliers:** Ensure there is adherence to agreed terms and conditions with suppliers and there are no negative procurement outcomes due to payment delays to suppliers that could result in supplier inefficiencies and higher prices being charged as a means of cushioning the financial effect of payment delays.
- **Supplier Performance Monitoring:** Ensure all accredited suppliers perform as contracted and make sure they do not retain their accreditation if they fail to perform as per their contracted obligations.
- **Effective and Efficient Warehouse Layout and Space Utilisation:** Ensure that the warehouse layout facilitates the efficient and effective movement, inwards and outwards, and, safe, quality assured, accountable storage of goods that are stored in a warehouse. The warehouse lay out refers to the maximisation of the floor area as well as the height (therefore volume) of a warehouse.
- **Efficient Receiving of Goods:** Ensure the efficient receiving of goods consistent with terms and conditions in the relevant purchase orders in the warehouse and move goods to a holding area as soon as possible and to minimise any inconvenience and/or delays to shippers who deliver goods.
- **Safe, Effective and Efficient Storage of Goods:** Ensure that warehouse space (floor area, height of the warehouse, therefore the volume, pallet racks, shelving space etc) is maximised and goods stored in such a manner that storage, picking and movement to a packing area is carried out maximising the efficiency of the labour force while safe handling of goods is done according to NDOH occupational health and safety requirements.
- **An Effective and Efficient Picking and Packing Operation:** Ensure that the picking and packing operation in the warehouse is planned, methodical, efficient and carried out by maximising the labour input, and is done accurately, under supervision and all stock picked and packed are in accordance with the relevant picking slips generated by the LMIS (mSupply), and all pickings confirmed in the LMIS (mSupply) with the least possible delay.
- **An Effective and Efficient Distribution Operation:** Ensure that distribution from the warehouse to health facilities occurs as per pre-arranged distribution schedules, at the tendered prices, and there is an accounting process to certify supplies have been delivered to the intended locations identified in the delivery schedules is carried out by distribution contractors chosen through a competitive tender process managed by the procurement entity (MSPD) or in provinces, by the PHAs or via provincial contracts arranged by the PHAs via a competitive tender process.
- **Effective And Efficient Stock Management:** Ensure stock levels are kept between MINIMUM and MAXIMUM levels determined by the procurement entity (MSPD) for all drugs and medical supply items in the warehouse, expiry prior to use is avoided, all transactions (incoming goods and outgoing goods) are accurately recorded in the LMIS (mSupply) using duly authorised documentation which are accountable and auditable documents and regular stock takes (monthly-cycle counting, and bi-annual stock counting) are carried out, any variations explained through a stock reconciliation process which requires approval from the senior executive authorised by the Secretary Health prior to any stock adjustments being done in the LMIS (mSupply).

HEALTH INFRASTRUCTURE: PLANNING, DESIGN, DEVELOPMENT, COMMISSIONING, OPERATION, MAINTENANCE AND DISPOSAL OF HEALTH FACILITIES INFRASTRUCTURE AND MEDICAL TECHNOLOGY MINIMUM STANDARD FOR HEALTH SERVICES

Health Facility Infrastructure and Medical Technology complexities increases with clinical and public health services complexities as delineated within 'Levels of care and health services'.

Health facilities infrastructure and medical technology planning, design, development, commissioning, facility management and assets disposal is informed by clinical and public health services requirements of the six (6) levels of care.

It is expected that the information herein will be of interest and value to Provincial Health Authorities (PHA) health service providers, funding bodies, architectural and engineering consultants and members of the construction industry involved in the design and delivery of new health service infrastructure in PNG.

The translation of the clinical and public health services into the health service facility infrastructure and medical technology is undertaken through clinical services planning which encompasses elements of clinical and public health and support services requirements.

Health facility design and medical technology standard guides the health facility infrastructure design and equipment to meet the defined clinical and public health services of a designated geographical and catchment area.

Health facility design standards comprises of architectural, engineering, environment, medical equipment and general furniture and fittings which are aligned to the Australasian and International Health Facility Guidelines.

Medical Equipment is regulated by International Medical Equipment/Devices regulations for safety and efficacy.

National Referral Specialist Tertiary and Teaching Hospital and all PHAs in collaboration with their respective health partners, are now required to:

- ensure that a 5-year health infrastructure and maintenance plan is developed and implemented, in conjunction with all relevant key stakeholders (eg: DDAs, Provincial Administrations, Local MPs), to support the PHA's 5yr strategic health services development plan;
- ensure that all key stakeholders (eg: DDAs, Provincial Administrations, Local MPs, etc) are aware of the NHSS health facility design standards and relevant health facility

accreditation approval process; and

- **report quarterly to the Board, through the CEO, on progress achievements of all new and/or refurbishments of health facilities for the organization.**

The planning, design, development and commissioning of PHA and healthcare organizations health facilities infrastructure and medical technology is undertaken through consultation with key stakeholders and in compliance with regulatory requirements.

NDOH Level One – Level Four Health Facility Infrastructure Standards, Design and Equipment Requirements provide the minimum standards to build and equip urban and rural health services and provide an outline of functional areas and equipment requirements.

Commissioning of new health facility and medical technology aims at ensuring that users are trained on the operation, use and maintenance of the assets.

Ongoing facility management ensures that PHA health facility infrastructure and equipment is kept functional and safe for patient, staff and guardians during operation.

FINANCIAL MANAGEMENT MINIMUM STANDARD FOR HEALTH SERVICES

Improving Financial Resource Management within health sector

In light of the recent fiscal crises faced by the PNG Government and in an effort to support Government's requirement to stabilize the economy and national budget to maintain macro-fiscal discipline in the medium-term, the health sector is going to face many challenges in terms of fiscal shocks, potential delays in the issue of warrants and regular cash shortfalls or limited disbursements throughout each fiscal year.

It is paramount that all levels of the PNG health system strengthen their financial mechanisms and processes by which they access and utilize public resources from several pools at national, provincial and district levels. This will also improve accountability for the delivery of health services across all levels of health services.

NHSS Public Financial Management (PFM) Summary:

- Improved management of resources to deliver health standards and priorities
- Better mobilization and utilization of public (and/or) other sources of funding
- Improved linkages between budgets and their objectives
- Improved cash flow predictability for budget execution across all levels
- Enhanced credibility of PHAs in delivering health services
- Improved internal controls

DEMAND FOR GREATER ACCOUNTABILITY AND TRANSPARENCY IN THE USAGE OF PUBLIC FUNDS ACROSS THE HEALTH SYSTEM

Effective public financial management is not an end in itself, but a means to ensure that health service delivery at all levels is effective.

NDOH as the lead agency and steward for the health sector will be required to focus on intergovernmental coordination and consolidation of efforts on health system strengthening, better health service delivery, improved acquisition and allocation of financial and other resources, both to the sector and within and across the sector.

NDoH will also facilitate a sector-wide approach to ensuring compliance with PFM legislation and guidelines, including major Public Financial Management (PFM) reforms.

PHA Boards, CEOs, Executive Directors and Corporate Services staff are now required to enforce accountability and transparency on the usage and delivery of all public funds across all health service levels, including with and within health partner organisations.

Given the medium-term risks to securing fiscal space across the PNG budget and having funds disbursed against those appropriations in a timely manner, it is up to the management of each individual PHA (the CEO, Senior Executive Management and Corporate Services Division) to influence

the fiscal allocation decisions during the budget formulation phase and cash disbursement decisions during the execution phase that are made by several different parties working within the PFM environment.

In the next few years, learning to '*do more with less*' may help the health system to adapt better to growing fiscal constraints and limited funding flows, as well as demonstrating to Central Ministries, Provincial and District Governments that as they regain fiscal stability, the health sector will be one of the first sectors to be ready to absorb and mobilize additional funds effectively and deliver better health outcomes across all levels.

HEALTH INFORMATION MANAGEMENT MINIMUM STANDARD FOR HEALTH SERVICES

Improving Health Information and Information Communication Technology (ICT) For Decision Making

At each level of care and service, including PHA Board, CEO & Executive Management teams, there is to be an understanding of the importance of data & information, and its use in evidence-based decision-making, in order to assist with managing & leading both the PHA and the Health Sector more effectively and efficiently.

A reliable health information system is essential to the effective functioning of the PNG health system and determining the ongoing health status of a population. It is therefore important to have accurate and timely information for:

- Monitoring & Evaluating patient care & preventative health programs
- Enhancing service efficiency and effectiveness, and continuous quality services improvements
- Ensuring adequate resource (money, people, equipment, facilities and services) allocations

The PHA Board requires relevant and quality information to inform its accountability requirements, as well as ensuring that the CEO and Executive Directors can use relevant and up-to-date information for making transparent and evidence-based decisions for safe quality health services and resources, and to support a PHAs annual report and associated financial statements.

National Referral Specialist Tertiary and Teaching Hospital and all Provincial Health Authorities (PHAs) are now responsible to maintain, monitor & evaluate the volume & quality of all health services and all health programs delivered at all levels of care, through regularly observing health information to inform about service efficiency and effectiveness, and using health information as the basis of evidence-based quality service improvements, and for justifications for adequate resource decision-making.

The NDOH also requires at a minimum, data and information to support a health sector-wide monitoring and evaluating framework and reports against national health indicators that monitor the performance of individual PHA health services and their achievement of national targets against the National Health Plan.

KEY PERFORMANCE INDICATORS

The '*quality of health care*' is on the agenda, and therefore, assessing the quality of care is to become increasingly important to all PHAs, health partners and other health service providers, regulators, and patient/consumers of health care.

In recent years, interest has been growing in evidence-based medicine and a focus on the cost-effectiveness of health care in producing health results/outcomes.

Key Performance Indicators (KPIs) for outcome measurement allows the quality and safety of health care services to be measured.

This assessment can be done by creating quality indicators that describe the performance that should occur for a particular type of patient and/or the related health outcomes, and then evaluating whether patients' care is consistent with the indicators based on evidence-based standards of care.

KPIs can be defined in several different ways:

- as measures that assess a particular health care process, procedure or outcome;
- as quantitative measures that can be used to monitor and evaluate the quality of important governance, management, clinical, and support functions that affect patient outcomes; and
- as measurement tools, screens, or flags that are used as guides to monitor, evaluate, and improve the quality and safety of patient care, clinical support services, and organizational functions that affect patient outcomes.

Key Performance Indicators (KPIs) provide a quantitative basis for clinicians, health service staff, organizational managers and planners aiming to achieve improvement in patient health care processes provided.

The use of KPIs enables health professionals and PHA and healthcare organizations to monitor and evaluate what happens to patients as a consequence of how well health professionals and organizational systems function to provide for the needs of patients.

Key Performance Indicators are based on established standards of care. Indicators are, however, not a direct measure of quality, because quality is multidimensional, understanding quality requires many different measures.

Health and other Information to support KPIs are derived from a variety of sources including through eNHIS, medical records, registers of births, surgical operations & deaths & from daily management activity reports of the various hospital departments, health facilities and corporate service functional areas.

Each PHA will have their own unique set of circumstances and environments they work within and prevalence rates vary based on those circumstances.

When reviewing the data for benchmarking and decision making at a strategic level, it is important that each PHA population demographics is taken into account.

Most of these data sources already exist within both PHAs and NDoH, and it is now required that regular auditing of those data sources is undertaken by both PHAs and NDOH to ascertain:

- **collection timeframes;**
- **accuracy of the data collected;**
- **that all relevant health services clinicians, managers, health service staff are actually collecting and recording the required health and other information in the formats required;**
- **adequacy of data storage and back-up of data (both on-site and off-site); and**
- **use of data for evidence-based decision-making and reporting purposes.**

PHA Health Information Officers (PHIOs) are required to take the lead to coordinate monthly reports working with PHA Hospital Medical Record Officers (MROs), with all District Health Managers (DHMs) and Rural Health Facility OICs, as well as corporate services staff.

The PHA Health Information, Statistics, Medical Records Management & ICT Advisory Committee reviews, analysis monthly reports & trends, & the findings used to initiate & monitor quality improvements.

ELECTRONIC NATIONAL HEALTH INFORMATION SYSTEM (eNHIS)

The PNG health sector's Electronic National Health Information System (eNHIS) is an integrated patient & health system reporting software package designed to:

- capture information relevant to the management of the health sector in Papua New Guinea; and
- provide reports, statistics & maps to support sector management and health services delivery.

The eNHIS encompasses the following modules:

- computer tablet-based data collection & reporting;
- web-based eNHIS access module; and
- resource library of guidelines & mapping modules.

The computer tablet is provided to Health Centres, District & Provincial Hospitals to capture patient activity data & provide feedback on health facility performance.

The tablets have the following attributes:

- Multi-password protection (login & OIC passwords);
- Outpatient data collection;
- Inpatient discharge recording;
- Malaria registers;
- Leprosy Registers;
- TB registers & 1st & 2nd line patient management;

- Document Library of resource materials & dictionary of health terminology;
- Public Health Bulletin & feedback to HC portals;
- Drug Stock out reporting;
- Inventory reporting;
- Contact reporting; and
- Daily, weekly, monthly, quarterly, 6-monthly and annual reporting.

The computer tablet network is designed to be scalable ie: more than one tablet can be deployed at a health facility depending on its size. This allows its use in different levels of the health sector eg: PHA Community Health Post, Health Centres, Urban Clinics, District Hospitals & Provincial Hospitals.

The Public Health Bulletin on health facility computer tablets allows relevant information to be sent back to the tablet from the PHA's PHIO. This includes:

- PHA Feedback, advice and instructions from the CEO or Executive Directors;
- Monthly reporting that is specific to a particular Health Facility or District; and
- Notifications from the NDoH concerning current issues that are of high national importance.

Messages or notifications can be sent back health facility computer tablets by using the "Messages/Email dashboard".

The eNHIS generates approximately 35 kinds of reports, including Program reports & inpatient reports. SPAR reports can now be automatically generated & downloaded for inclusion within management reports or into excel spreadsheets for further analysis.

Health Program reports can also now be easily downloaded and distributed to responsible health services managers and staff to enable the monitoring, evaluating, and improving of program performance.

MONTHLY REPORTING

The following reports are to facilitate consistent quarterly reporting to PHA Boards and the NDOH, for progressive compilation of PHA Annual Reports.

The National Referral Specialist Tertiary and Teaching Hospital, PHA's Executive Director's Curative Health Services, Public Health Services and Corporate Services, are all individually and collectively, now responsible to ensure that all persons completing monthly and quarterly reports fulfil their respective legislative responsibilities.

If utilised correctly, Monthly and Quarterly Reporting addressing the selected indicators meets both PHA and NDoH reporting obligations, and provides the necessary information to allow well-informed management, clinical and corporate decisions.

The integrity of any aggregated reporting at PHA & NDOH level is dependent upon the use of consistent definitions & data collection methods. The use of reported information for comparative

purposes is only valid when it has been collected using the same definitions & methods at each source.

All PHA and NDOH Officers are required to complete Monthly Reports are to be familiar with the following information.

SECTOR PERFORMANCE ANNUAL REPORT (SPAR)

The **Sector Performance Annual Report (SPAR)** is based on eNHIS and is linked to the NDOH M&E framework, that specifies core health indicators; data sources, analysis & use; and inclusive transparent mechanisms for review & further actioning.

A tested, relevant, balanced & parsimonious (sparing) set of core indicators is one critical element that contributes to the overall strengthening of the health system & its accountability.

The overall performance of the PNG health sector is based on the twenty-eight (28) health indicators through the SPAR annual report.

The performance of each PHA province is assessed against each of the selected indicators. If there is improvement (of more than 2% from year to year), the PHA province is assigned a score of one for that indicator. If there is no change, a score of zero is applied, and if performance has decreased (by more than 2%) a score of -1 is applied.

These scores are accumulated to provide the basis for ranking the individual PHA provinces according to improvement across all provinces. Similarly, comparisons of every individual District within each PHA province is undertaken.

The performance has been sub-analysed according to the type of indicator - outcome, output, process or input. If a PHA province has improved inputs or processes, but has not improved its outputs or outcomes, then there is a need to consider the management approaches to be taken.

An extensive set of KPI and Clinical Indicators are for further debate, and review over the next coming years.

INDICATORS WITHIN CLINICAL SPECIALTY AREAS

The aim is to reduce the indicator reporting burden for PHA & feedback from PHAs Senior Manager to healthcare providers & from the NDOH to PHA senior executive team.

There are numerous data sourced from a broad range of clinical specialty areas. Any Clinical Indicator Reporting needs to provide key points on significant trends, & differences for a broad range of Clinical Indicators. Such a report also includes commentary by healthcare professionals within the respective healthcare specialty to provide context to the complex & ever-changing healthcare environment & provide insight for the potential to improve quality & safety within their health services.

The indicators are a set of worldwide indicators adapted to the PNG healthcare environment to provide valuable knowledge of our healthcare system.

In order to enable performance assessment and benchmarking, and to support improvement in health outcomes which are important to patients, ideal clinical indicators are well defined & easy to collect, evidence-based, specific, sensitive, valid & reliable. Trends in the rates over time are reported with statistical significance, & the data are displayed in graphs.

The Indicator data is provided by clinical or medical records officers familiar with the ICD 10 & eNHIS definitions & data dictionaries. The collection of this information is reflected in the National Quality Standards relating to Governance, Public Health & Clinical Care & Management Information Systems.

HUMAN RESOURCE UTILISATION AND REPORTING

Health workforce related costs including salaries, wages, overtime, leave fares & accommodation rental charges account for up to 90% of some PHAs' budgets, and as such, demand special attention from the PHA CEO and Executive Management.

Whilst it might be argued that the present system of appropriation for labour-related costs does not provide incentive for PHA Executive Teams to manage their level of vacancies, unattachments, overtime payments & staff productivity it nevertheless is an obligation of the PHA Board to ensure that the CEO and Executive Management to do so.

Boards, CEOs and Executive Teams are now required to closely scrutinise the status & performance of all cadres of clinical health care provider staff and corporate support staff, delivering services across the PHA.

There is increasing evidence to show that the failure to effectively manage staffing levels & productivity is a determinant in the refusal by the Department of Personnel Management/ Department of Finance to allow PHA vacancies to be filled.

Some health services staff delivering clinical services at particular Hospitals, or Urban-Rural Health Services, or Public Health programs may not necessarily appear on the payroll of that particular health service or health program (e.g. Senior Medical Officers paid directly from the National Department of Health's budget allocation) however, they do form part of the available human resources to deliver services at that PHA's health facility, and their presence or otherwise is acknowledged in the Monthly Report, if inter-hospital comparisons of productivity/output are to be valid.

The definitions used to categorise and classify staff and to report their status are those used in the PS General Orders and generally accepted by the Department of Personnel Management/Public Service Commission.

FINANCIAL REPORTING

If the approved budget appropriations for each PHA are tied to the annual implementation plan across the PHA, and if all payment officers properly record expenditures and commitments by filling in the correct information in payment vouchers and these are processed properly and entered back into either IFMS or PGAS under the correct expenditure vote, then each PHA can develop a much better understanding of:

- What level of warrant authority has been granted.
- What level of resources have been provided and from which sources (whether they be direct grants for the operational budget or development budget, or whether they be from internally generated revenue from provincial government, or fees charged for medical and other services, or donor funds, or other types of grants).
- How it has been distributed through the CEO and PHA Finance and Budget Committee (and Section 32s) towards delivering the PHA's AIP.
- How health facilities have used those funds or reported against the cash disbursed.
- What has been achieved (based on financial information that is linked to SPAR indicators, key activities and programme areas).

The monthly reviews of the warrants authorized, cash released and spending against those warrants can then be amalgamated to develop three-month expenditure and revenue reviews against HR information and the eNHIS indicators.

These reviews must be substantiated with quarterly reports that form the basis for the Annual Management Report of the PHA, to ensure that financial performance reviews are being tracked against the delivery of the PHA's targets and objectives of the year.

Monthly Status Reports on Actual spending and commitments can be structured as follows:

- By cost centre i.e., the provincial hospital and district hospital or hospitals (under curative health), district health facilities (under the public health division), facility levels 1 to 3 and Catholic and Christian health services under 'provincial and rural health services' as well as Corporate Services and Executive Management Services.
- By vote number, including warrant authorization from the previous month, plus the current month to give a total amount allocated to a particular expenditure vote. Provide a summary of the commitments against the previous months' warrant authority and cash disbursements that have to be included in this month's commitments, plus current commitments which will give 'total commitments', against which total payments can be compared to determine if the commitments are being met.
- By programme area of spending and the economic items for various 'types' of spending under each agree programme area and activity (as per the health CoA).

The planned spending against the AIPs and actual spending against these AIPs can be compared to review:

- i. Where the commitment of funds is close to 100% exhausted.
- ii. Where commitments have been adjusted or revised due to a change of work plan or activities.
- iii. Whether there have been any 'irregular items of expenditure' or significant underspending against planned spending even with the availability of cash.
- iv. Any major planned areas of spending.
- v. Any expenditure items which were NOT budgeted for originally.
- vi. Any purchase of assets, medical equipment etc.
- vii. By specific public investment projects (PIPs).
- viii. Progression of works against the agreed contract.
- ix. Any major variations
- x. Any emerging issues and challenges to implementation
- xi. Any revision to the scope of works for the year.
- xii. By service delivery and goods contracts that were awarded, signed and have commenced.

CLINICAL GUIDELINES

Clinical guidelines are designed to support the decision-making processes in patient care. Although not within the National Health Service Standards, they are important links to improve safe, quality patient care.

The content of a clinical guideline is based on a systematic review of clinical evidence - the main source for evidence-based care.

The movement towards evidence-based healthcare has been gaining ground quickly over the past few years, motivated by clinicians, and management concerned about quality, consistency and costs.

Clinical guidelines based on standardised best practice have been shown to be capable of supporting improvements in quality and consistency in healthcare.

Many have been developed and disseminated in Papua New Guinea, though the process is time and resource consuming.

The purpose of the clinical guidelines is to:

- Describe appropriate care based on the best available scientific evidence and broad consensus;
- Reduce inappropriate variation in practice;
- Provide a more rational basis for referral;
- Provide a focus for continuing education;
- Promote efficient use of resources; and
- Focus on quality control, including audit.

Clinical protocols can be seen as more specific than guidelines. Protocols provide "*a comprehensive set of rigid criteria outlining the management steps for a single clinical condition or aspects of organisation*".¹ Currently Papua New Guinea has many outdated treatment manuals /clinical guidelines, and a small number of up-to-date guidelines i.e. Infection Prevention Guideline. Furthermore, many manuals or guidelines are not endorsed and some require further editing.

Over time, the National Department of Health via the Medical Standards Division and clinical leaders will review, update and ensure broad dissemination and monitoring of relevant clinical guidelines /treatment manuals.

Annex Two outlines the list of the National Department of Health guidelines, and manuals.

¹ Medical Guideline; Wikipedia Encyclopaedia, 2001.

NATIONAL QUALITY STANDARDS AND COMPULSORY ACCREDITATION PROGRAM FOR HEALTH SERVICES NHSS VOLUME 2

The **National Quality Standards for Health Services and the Health Service Accreditation Program** is an important component of the National Health Services Standards and a tool that can assist in strengthening health services including primary health care, public health, rural health and hospitals to improve service delivery.

The National Quality Standards (NQS) for Health Services and the Health Service Accreditation program provides healthcare services with an organisation-wide framework to deliver a consumer-centred service.

Volume Two of the NHSS provides a framework, commitment and process for the use of PNG national healthcare standards to achieve continuous improvement, *'to strengthen safe, quality health care by continuously advancing standards and education nationally'*.

The 2011 NHSS Quality Standards were initially based on the 1997 hospital standards and then expanded to include rural health services.

This 2nd NHSS edition builds on the NHSS Volume 2 Quality Standards and addresses the broader National Quality Standards and Compulsory Accreditation Program based on a Package of Integrated Health Services Standards for Provincial Health Authorities.

The National Quality Standards processes support quality improvements in healthcare and commitment to ensuring these National Health Services Standards (NHSS) are current, comprehensive and in alignment with the National Health Plan, best practice and that furthermore, the standards are able to be assessed in a way that generates useful, accurate information to improve safety and quality for our patients and the broader community.

Each PHA will select from within each level of care across their districts and include health services and health facilities (government, church health service providers and non-government organisations), priority public health programs and partnering non-government organisations that will be assessed against the National Quality Standards for Accreditation.

The key components of the National Quality Standards (NQS) & Accreditation are:

- National Quality Standards a PHA (including church health services, NGOs and private sector organisations) work to achieve,
- Annual self-assessment to evaluate performance against set NQS,
- NDOH and across PHA Peer Surveyors assistance and guidance with organisations self-assessment,
- Peer Surveyors (a team of accreditation trained surveyors) provide an independent assessment of the PHA organisation's performance against the NQS,
- Quality improvement process undertaken by PHA – healthcare organisations to address the recommendations from the onsite surveys,
- Accreditation based on Survey Report and approval of National Health Board (NHB).

The standards comprise of four (4) core functions, a series of standards, criteria and elements, arranged under graded ratings and risks that reflect increasing maturity of a PHA – healthcare organisation's quality improvement activities.

Effective use of the National Quality Standards (NQS) for Health Services within Papua New Guinea requires an integrated organisational approach to quality improvement. This approach will assist PHA and healthcare organisations to:

- Focus on patients/consumers/customers,
- Develop strong leadership,
- Build effective systems through ongoing improvement,
- Measure and evaluate outcomes, and
- Strive for best practice.

The **Core Functions** in the NHSS 2nd edition sets out against the National Quality Standards enable PHAs – healthcare organisations to achieve quality improvements against the standards and address:

1. Organisational Governance and Health System Strengthening for a PHA – healthcare organisations,
2. Quality & Safety in Integrated Health Service Delivery,
3. Essential Public Health Services, and
4. Essential Clinical Care.

ACCREDITATION

Accreditation is the status obtained by a PHA – Healthcare organisation after a successful peer survey external evaluation facilitated by the NDOH to assess whether the PHA – Healthcare organisation meets applicable predetermined and published National Standards.

Accreditation is one aspect of performance measurement-requirements set within the NHP & NHSS. The focus of accreditation programs is to provide a framework for continuous improvement. It is about establishing a structure and processes that allow quality and safety to proliferate.

PHA – Healthcare Organisations that successfully achieve an accreditation status endorsed by the National Health Board and awarded a full four years accreditation receive country wide recognition by the Minister of Health and official acknowledgement of the PHA – Healthcare organisation's commitment to continuous quality improvement. The Minister for Health will sign and present a Certificate of Accreditation promoting quality. PHAs – Healthcare Organisations celebrate the achievements of accreditation.

The **NHSS Volume 2 National Quality Standards and Accreditation Program for Health Services in Papua New Guinea** is intended to guide all Provincial Health Authorities in the use of the National Quality Standards for Health Services in Papua New Guinea and in the accreditation process, i.e. PHA

Boards of Governance, Chief Executive Officers and Senior Management Teams, Hospital and Health Service Managers, National Department of Health Personnel and the Accreditation Peer Surveyors.

NATIONAL DESIGN STANDARDS FOR HEALTH SERVICES NHSS VOLUME 3

Volume Three of the National Health Services Standards (NHSS) 2021-2030 is to inform all stakeholders involved in the **planning, design, development, commissioning, operation, maintenance and disposal of health facilities infrastructure and medical technology for the public sector in PNG.**

It is expected that the information herein will be of interest and value to Provincial Health Authorities (PHA) health service providers, funding bodies, architectural and engineering consultants and members of the construction industry involved in the design and delivery of new health service infrastructure in PNG.

Most of the facilities from which health services are delivered in PNG at this time are in poor condition and will be systematically replaced through the implementation of the Government's 'Vision 2050' and the National Department of Health's (NDOH) National Health Plan 2011-2020 or its successor.

It is therefore timely that the NDOH disseminates standards for the design and construction of new health infrastructure to ensure that the capital works program consistently delivers facilities which provide value for money, a sustainable asset and a modern clinical environment to support the work of our health professionals.

Compliance with these Design Standards will also:

- Ensure new healthcare facilities provide a clinically safe, culturally appropriate, efficient and comfortable physical environment for clients, visitors and staff members;
- Ensure standardisation of facilities provide familiar environments for staff moving across the sector; and
- Ensure, within reason, the equitable provision of amenities over time across the sector.

CONSULTATION ON NATIONAL HEALTH SERVICE STANDARDS

Stakeholder alignment is critical to successful implementation of the standards and in order to account for this, the NDoH Medical Standard Division supported the formulation of the National Health Service Standards (NHSS) Technical Working Group (TWG) and SEM endorsed, enabling an ongoing consultation process.

The NHSS TWG continues to meet to review progress in the implementation of the NHSS and to:

- Provide an overview, background and context on gaps, challenges and achievements of the NHSS in health service delivery.
- Gather credible evidence i.e., data sources, quantitative questionnaires/survey, existing research, literature reviews.

Consider:

- PNG capacity to achieve Universal Health Coverage, and the Sustainable Development Goals and bridge the access gaps in primary health care service delivery.
- Strengthening NHSS in improvements in maternal and newborn care in line with the PNG Maternal Health Strategy.
- Strengthening PHAs formalisation of an integrated health system which includes Patient Referral Systems and guidelines.
- Reviewing current workforce, productivity, throughput and population needs in regard to skill mix, workforce requirements and continuous professional development to ensure safe quality patient 24-hour care, integrated outreach, supportive supervision and public health program activities for each level of health service delivery.
- Reviewing gaps in infrastructure and equipment based on each level of health service delivery functionality, preventative maintenance programs and management of the health facility compounds.
- Consider the barriers to PHA good governance and efficiency and effectiveness in supporting the implementation of NHSS.
 - Address compliance to NHSS including continuous quality improvement and accreditation processes,
 - Address supportive supervision, and
 - Address continuing professional development and health worker training.
 - Ensure a process for monitoring, evaluation and reporting.

IMPLEMENTATION AND MONITORING OF NATIONAL HEALTH SERVICE STANDARDS

The marketing of and roll out of the National Health Service Standards to major stakeholders is an essential component of implementation. The National Department of Health's Divisions and Provincial Health Authority Boards and CEOs play a key part in this implementation.

The National Department of Health Medical Standards Division supported by the Public Health Division will take a coordination role for a team of trained peer surveyors to monitor compliance with the National Health Service Standards. The monitoring of these standards will be achieved through:

- Delineation of PHA - healthcare organizations levels of care and health services;
- Health services accreditation surveys against the National Quality Standards;
- Operational review; and
- Medical Board licensing processes.

SUMMARY

The 1997 NHAA requires the National Department of Health to develop a single health policy and National Health Standards.

The National Department of Health has a mandate to establish policy, in this instance, the National Health Service Standards. These Standards promote the best possible quality of health care and continuous quality improvement of health services.

The endorsed National Health Service Standards will guide health service delivery to achieve safe levels of quality health care in Papua New Guinea.

ANNEXURES

ANNEX 1:

PNG NATIONAL HEALTH SERVICE STANDARDS TECHNICAL WORKING GROUP TERMS OF REFERENCE

NDOH Division:	Medical Standards & Public Health Divisions supported by Health Services Sector Development Project under the Direction of the Secretary for Health.
Technical Working Group:	PNG NHSS Technical Working Group
Duration:	Time limited
Key Partners:	HR Workforce, Selected PHA CEOs, Partners
Reporting Structure	NDoH Deputy Secretary NHP & Corporate Services, via CMO/Executive Manager, Medical Standards Division
Keywords	sustainable development goals, health-related indicators, achieving universal health coverage, effective, safe, people-centred care, adherence to clinical practice guidelines, model of care, culture of improvement, efficiency and effectiveness, productive health workforce, community engagement, effective partnerships, clinical supervision & training.

Health Services Delivery reflects a decentralised integrated health care system and a deliberate focus on safe quality patient care, and effective public health interventions that enhances community. The NHP Plan 2021-2030 – ‘LEAVE NO ONE BEHIND’.

PNG REVIEW OF NATIONAL HEALTH SERVICE STANDARDS NHSS TECHNICAL WORKING GROUP TERMS OF REFERENCE

AUTHORITY

The National Department of Health, Senior Executive Management Committee (SEM) has approved HSSDP to support Medical Standards and Public Health Divisions to review the National Health Service Standards. The NHSS Technical Working Group is established to advise the NDoH Deputy Secretaries, via CMO/Executive Manager, Medical Standards on the review and further directions and recommendations in line with the Strategic Directions and proposed outputs of the National Health Plan 2021-2030.

PURPOSE

The NHSS Technical Working Group purpose is to review and update the NHSS and align with the development of the new NHP 2021-2030 reflecting GOPNG health agenda.

CHALLENGES

Challenges in strengthening the PNG health system include (i) fiscal constraints, fragmented funding streams, and weak PFM, (ii) frequent medical supplies stock outs compromising clinical and health outcomes, (iii) an ageing workforce, (iv) inconsistent standards of governance, and management including effective use of information, (v) inconsistent health seeking behaviours by community members (e.g., for immunization), (vi)

gender inequality contributing to poor health indicators, (vii) health infrastructure decaying due to lack of investments including maintenance, (viii) a misalignment between resources and accountability, (viii) inconsistency in the equitable distribution of funds and (ix) insufficient clinical supervision and training.

In response to these challenges, the Government of PNG (GOPNG, the government) is (i) aligning its national development plans to achieve the Health Sustainable Development Goals (SDGs), (ii) implementing financial management reforms, (iii) supporting the removal of user fees for primary health care, and (iv) subsidizing selected specialist health services. The government plan is that health system decentralization and integration will be completed by July 2019, locally governed and managed by Provincial Health Authorities (PHAs).²³ A March 2015 independent review found that this PHA-based, decentralized model contributes to improved health outcomes in PNG.⁴

In preparation for the Review of the NHSS and in line with the development of the National Health Plan 2021-2030, there are a series of discussion papers on Universal Health Coverage (UHC) and Essential Packages of Care (EPOC) that enables debate and discussion.

SUSTAINABLE DEVELOPMENT GOALS (SDGS) IMPROVE HEALTH OUTCOMES OVER THE LONG PERIOD.

One goal (SDG3) specifically sets out to “ensure healthy lives and promote well-being for all at all ages.” Its 13 targets build on progress made on the MDGs and reflect a new focus on noncommunicable diseases and the achievement of universal health coverage. The linchpin of development in health and reflecting the SDGs is a strong focus on equity and reaching the poorest, most disadvantaged people everywhere.

Key ingredients for success included a doubling in global funding for health, the creation of new funding mechanisms and partnerships, and the critical role of civil society in tackling diseases such as HIV/AIDS and scale-up of new interventions such as antiretroviral therapy for HIV treatment and insecticide-treated bed nets to prevent malaria. Of the 41 health-related SDG indicators, the following are noted: mortality rate due to road injuries, prevalence of intimate partner violence among women, maternal mortality ratio, and incidence rate of new HIV infections. Key areas outlined in the health SDGs:

- reproductive, maternal, newborn, child and adolescent health;
- infectious diseases including HIV, tuberculosis, malaria, hepatitis and neglected tropical diseases;
- noncommunicable diseases (NCDs) including heart disease, cancer and diabetes;
- mental health and substance use including narcotics and harmful use of alcohol;
- injuries and violence; and
- universal health coverage.

ACHIEVING UNIVERSAL HEALTH COVERAGE (UHC)⁵

All people and communities, everywhere in the world, should have access to the high-quality health services they need – promotive, preventive, curative, rehabilitative, or palliative – without facing financial hardship. Quality of care, especially patient safety, is essential to creating trust in health services. It is also key to global health security, which starts with local health security, and in turn depends on high-quality frontline health services. Quality health services not only prevent human suffering and ensure healthier societies, they also ensure better human capital and healthier economies.

² Sector Development Program, Papua New Guinea: Health Services Sector Development Program, March 2018.

³ Government of Papua New Guinea. 2017. *100 Day Plan*. Port Moresby; Government of Papua New Guinea. 2013. *Free Primary Healthcare and Subsidized Service Policy*. Port Moresby; Government of Papua New Guinea. 2007. *Provincial Health Authorities Act 2007*. Port Moresby.

⁴ Government of Papua New Guinea. 2015. *Independent Review. Provincial Health Authority Management and Structures*. Port Moresby.

⁵ Universal Health Coverage, The World Bank, Washington, DC.

Optimal health care cannot be delivered by simply ensuring coexistence of infrastructure, medical supplies and health care providers. Improvement in health care delivery requires a deliberate focus on quality of health services, which involves providing effective, safe, people-centred care that is timely, equitable, integrated and efficient. Quality of care is the degree to which health services for individuals and populations increase the likelihood of desired health outcomes and are consistent with current professional knowledge. A systematic review has shown that suboptimal clinical practice is common in private, public and church run primary health care facilities in several low- and middle-income countries. Examples:

- Adherence to clinical practice guidelines- in low- and middle-income countries was below 50% resulting in low-quality antenatal and child health and deficient family planning.
- Service Delivery Indicators - in low- and middle-income countries showed significant variation in
 - service provider absenteeism (14.3–44.3%),
 - daily productivity of service providers (5.2–17.4 patients),
 - diagnostic accuracy (34–72.2%), and,
 - adherence to clinical guidelines (22–43.8%).

ESSENTIAL PACKAGES OF HEALTH SERVICES (EPHS)⁶

Packages of essential (or basic) health services have been formulated in several health sectors, for different reasons, not always explicit. The package inspired the PHC concept promoted by the Alma Ata Conference which stressed social justice and empowerment. The concept retains considerable appeal, because of its promise of explicit, evidence-based, rational priority setting. The purpose of the EPHS serves 4 basic purposes: expand the standardised primary packages of health services, provide equitable access to essential hospital services, strengthen the service delivery network and provide basis for operational plan development.

PACKAGES, PLATFORMS, AND POLICIES

‘Platforms’ are defined as logistically related delivery channels, a platform is the level of a health system at which interventions can be appropriately, effectively, and efficiently delivered. DCP3⁷ groups EUHC interventions within packages that can be delivered on different types of platforms. The following is suggested as the health service delivery platforms:

OVERARCHING PUBLIC HEALTH

Population-based strategies and health interventions across all platforms: this captures all population-based health services, such as mass media and social marketing of educational messages, as typically delivered by public health agencies.

LEVELS OF SERVICE

The NDOH SEM Team considers and approve that:

- The Aid Post Level 1 health service facility is maintained in the new NHP 2021 to 2030 based on the facts and considerations provided;
 - the name Aid Post is changed to *Health Post* in the new NHSS and NHP 2021 to 2030;
- Regional Hospital level 6 is done away with in the new NHP 2021 to 2030 based on the facts and considerations provided;
- The new NHSS will have six (6) levels of health service facilities and that the NHP 2021 to 2030 will be structured accordingly to reflect these levels;
- Provincial Health Authorities will define their health service delivery models based on provincial health service development plans;

⁶ Essential Package of Health Services: Primary Care: The Community Health System & Concepts and Strengthening Primary Health Care (PHC) Services, & Secondary & Tertiary Care: review of literature, WHO, Geneva.

⁷ UHC – Disease Control Priorities, Vol 9, 3rd edition.

- Every Provincial Health Authorities will develop their patient referral pathways (guidelines) in line with health service delivery models and provincial health service development plans.

KEY ISSUES

1. Nationally, the Department of Health (DOH) is shifting its focus and functions from direct management of service delivery to strategic management and coordination of policy, regulation, monitoring, and health sector budgeting, as the health system fully decentralizes.
2. Sub-nationally, health system strengthening will be through service delivery and implementing public health programs based on:
 - evidence-based planning,
 - corporate and clinical / public health governance,
 - leadership and management including financial management,
 - partnerships with district authorities, the private sector, NGOs and CHS/ CCHS, communities,
 - improved information systems and their effective use,
 - community health awareness raising and improved health seeking behaviours,
 - medical supplies distribution and
 - civil works for new health facilities for catchment populations, geography and disease burdens (based on health service planning).
3. Adherence and compliance with 2nd edition NHSS based on:
 - a. Achieving, within all PHAs, a culture of service and management improvement that is conducive to continuous improvement i.e., clinical governance standards, and guidelines (tool kits) for streamlined quality assurance, using evidence-based policies and standards across public and private sector health facilities.
 - b. Achieving PHA NHSS baseline assessments and completion of health facility reports by each PHA, for interpretation to align with levels of care / platforms and essential package of care.
 - c. Achieving 22 PHA Health Service Development Plans.
 - d. Investing resources into a formal compulsory accreditation process e.g., financial resources, trained accredited surveyors (who are skills and experience to assess PHAs health services – size, complexity, remoteness) and an increased focus on safe quality patient care and public health interventions (e.g., current accreditation process measures procedural correctness based on examining large volumes of documentation to demonstrate compliance).
 - e. Using timely standardised data collection and analyses to better inform assessment processes i.e., eNHIS Hospital and RHS data collection includes quality data assurance and up to date data entry.
 - f. Strengthening efficiency and effectiveness of service delivery through an agreed model of care that supports improvement and advocacy for patient-centred care, promoting a collaborative approach to health care management and decision-making.
 - g. Strengthening the public and private sector in service improvement which includes regulation and credentialing of health practitioners and health facility licencing, and registration.
 - h. Strengthening a matrix of strategic role statement (platforms & packages) focusing on clinical networks, clinical groups, clinical services, hospital and primary health care / public health service capabilities (the functional level of health service delivery) that provides health services a consistent language to describe health services & public health programs and acts as a tool for planning future service developments.
 - i. Considering health service delivery networks (clinical services & rural and remote networks) and networking together with patient referral patterns and patient flows, population clusters and transport infrastructure (road, sea and air).
 - j. Enabling a productive health workforce that is efficient and effective through: distribution, skilled and motivated cadres of health workers that meets the skill mix for service delivery. Health Workforce training and Lifelong learning through Continuous Professional Development (CPD) by supporting and

developing the capacity of preservice training institutions to deliver high-quality educational programs aligned with the latest evidence and national health priorities, as well as the delivery of CPD learning packages based on latest evidence and national health priorities.

- k. Strengthening pharmaceutical systems by timely access to quality assured affordable effective and safe pharmaceutical products and ensuring their appropriate use i.e., medicines, vaccines, medical gasses, diagnostic testing e.g. (PoCTs). This includes delivery of safe, effective, quality-assured medicines that meet international standards.
- l. Incorporating health infrastructure - health facility design standards and medical & non-medical / static plant equipment and asset management standards and strengthen a planned approach for procurement, planning, management, inventory, preventative maintenance and budgeting for medical and non-medical equipment based on role delineation of each level of health facility, adequately trained healthcare workers in use of and care of medical equipment and devices. Each level of health service requires essential and basic medical equipment, cold chain equipment, non-medical equipment, essential drugs, vaccines, supplies and medical gasses.
- m. Implementing supportive supervision and PHA score cards to measure, monitor and increase compliance with evidence-based health standards and guidelines, and improve the quality of health services.
- n. Strengthening community engagement and effective partnerships between communities and health services, changing health seeking behaviours of individuals and communities.

TERMS OF REFERENCE

The Technical Working Group Terms of Refence will advise the NDOH SEM on the following:

- Provide an overview, background and context on gaps, challenges and achievements of the NHSS in health service delivery.
- Gather credible evidence i.e., data sources, quantitative questionnaires/survey, existing research, literature reviews.
- Consider
 - PNG capacity to achieve Universal Health Coverage, and the Sustainable Development Goals and bridge the access gaps in primary health care service delivery.
 - Strengthening NHSS in improvements in maternal and newborn care in line with the PNG Maternal Health Strategy.
 - Strengthening PHAs formalisation of an integrated health system which includes Patient Referral Systems and guidelines.
 - Reviewing current workforce, productivity, throughput and population needs in regard to skill mix, workforce requirements and continuous professional development to ensure safe quality patient 24-hour care, integrated outreach, supportive supervision and public health program activities for each level of health service delivery.
 - Reviewing gaps in infrastructure and equipment based on each level of health service delivery functionality, preventative maintenance programs and management of the health facility compounds.
- Consider the barriers to PHA good governance and efficiency and effectiveness in supporting the implementation of NHSS.
 - Address compliance to NHSS including continuous quality improvement and accreditation processes,
 - Address supportive supervision, and
 - Address continuing professional development and health worker training.
- Communicate findings and develop key recommendations for the development of the next version of the PNG NHSS.

- Once key directions and recommendations are endorsed by NDoH SEM, develop and finalize NHSS and package of tools and learning materials to support implementation and compliance.
- Ensure a process for monitoring, evaluation and reporting.

ROLES AND RESPONSIBILITIES OF MEMBERS

- To work collegiately to ensure consensus is reached on recommendations.
- To keep all discussions and strategic directions at meetings confidential and recommendations are communicated to NDOH Sem.
- To be sufficiently informed prior to meetings regarding the agenda and any supporting documentation so as to ensure efficient use of time, and well-informed decision making.
- To ensure that conduct and activities of the Working Group will, at all times, be cognizant of the Laws of PNG and all relevant policies, procedures, standards and guidelines of the National Department of Health.

MEETING SCHEDULE

The NHSS Technical Working Group members will commit to meeting at least monthly to initially review then address NHSS options and strategies.

MEMBERSHIP

Membership of the NHSS Review Technical Working Group will include:

MEMBERSHIP	
NDOH	Deputy Secretary, National Health Plan & Corporate Services. Dr Goa Tau, or representative, CMO/Executive Managers Medical Standards (Chair). Dr. Sibauk Bieb, Executive Manager Public Health. Dr. Esorom Daoni, Public Health. Mr. Ken Wai, Executive Manager, or Mrs Agnes Pawiong, Strategic Policy. Dr. Edward Waramin, Division of Public Health / Maternal Health. Mr. Ambrose Kwaramb, Health Facility Branch. Dr. Varaga Laka, Workforce Standards & Accreditation. Mr. Dixon Dimiri, A.Manager, Workforce Standards & Accreditation. Mr. Vali Karo, A.Manager, Pharmaceutical Services, Medical Standards. Mr. Peter Pindan, Technical Adviser, Medical Standards. Mrs Margaret Asinimbu, Technical Advisor, Legal Services. Ms. Lina Wam, Technical Adviser, Human Resources.
PHAs & POMGH	Dr. Duncan Dobunaba A. CEO Oro PHA. Dr. James Amini, A.CEO Central PHA. Dr. James Kintwa, CEO Hela PHA. Dr. Paki Molumi, CEO Port Moresby General Hospital. Dr. Trevor Kelebi, Director Public Health, West Sepik PHA. Mrs. Christine Gawi, Hospital Manager, Modilon Hospital, Madang PHA. Mrs. Margareth Samei, CEO, Tabubil Hospital, Western PHA.
Medical Board / Nursing Council	Professor Nakapi Tefuarani, PNG Medical Board. Dr. Nina Joseph, Registrar PNG Nursing Council .
Clinical Chiefs	Dr. Cornelia Kilalang, A.Paediatrician. Dr. Dora Katal, Chief Radiologist. Dr. Hoxson Okti Poki, Chief Surgeon. Dr. Ligo Augerea, Chief Obstetrician Gynaecologist. Dr. Seluia Mabone, Chief Pathologist. Dr. Wesong Boko, Chief Physician.
CHS	Mr. Ulch Tapia, Secretary, Christian Health Services. Sr. Jadwiga Faliszek, Secretary, Catholic Health Services.

HSSDP	Mr. Rob Akers, Project Manager, RPHSDP/ HSSDP. Ms. Vicki Assenheim, Clinical Specialist Advisor. Mr. Peter Baran, HR & Organisational Development Specialist Advisor. Kel Browne, Social Safeguards & Gender Specialist.
Partners	Ms. Anna Maalsen, Coordinator Universal Health Coverage, Life course and Healthier Populations, Division of Health Systems, WHO.

Chairperson: The CMO/Executive Managers Medical Standards will be the Chairperson, and when unable to attend, the members in attendance will elect an Acting Chairperson from those present.

Secretariat: Mr. Maluo Magaru, Senior Technical Officer Hospital Standards and Operations, MSD, and when unable to attend, Mr. Jeffery Kaupa, Executive Support Officer, Medical Standards supported by Ms. Vicki Assenheim, Clinical Specialist Advisor, HSSDP.

Appointment and removal of members: Members of the Technical Working Group (including ex-officio members) are appointed by virtue of their substantive position within the NDoH/PHA. They will remain members whilst they occupy that substantive position or are the most appropriate nominee.

Working Group members represent their agencies and should represent their interests in relation to the 'NHSS and implementation for health service delivery'; membership is attached to the position in the agency represented rather than the individual and members are encouraged to arrange proxy representation if they are unable to attend. A member's position on the Technical Working Group will be declared vacant if the member is terminated from their substantive position; is absent for 3 consecutive meetings; and/or is deemed by the Working Group to have breached confidentiality.

The Technical Working Group may solicit the views and opinions key informants including, but not limited to DoH, PHAs, Clinical Chiefs, relevant development partners including Non-Government Organizations (NGOs) supporting service delivery. Extensive Consultation will be with officers within Medical Standards and Public Health Divisions of the DoH as well as with Provincial Health Authorities.

CONVENING OF MEETINGS

- Will be held monthly, with notice to be supplied a minimum of 1 week prior to the agreed date of the next meeting.
- Agenda, business paper and reports for consideration will be supplied by email 1 week prior to the next meeting.
- Meetings should NOT be cancelled because a member is unable to attend.
- Additional meetings may be convened by the Chair following written request from at least 2 other members and a week's notice must be provided to members prior to proceeding.
- Urgent issues can be considered by the members at the discretion of the Chair.
- Communication between Technical Working Group members outside of formal meetings will be by email, WhatsApp or phone.
- Members should respond to requests as soon as possible.
- Budget for Technical Working Group meetings (POM based) remains the responsibility of the NDoH HSSDP.

MINUTES OF MEETINGS

Will be the responsibility of the CMO/Executive Managers Medical Standards.

Action minutes of meetings will be circulated within 5 working days of a meeting.

Electronic copies of all agendas, business papers, reports and minutes will be archived at the HSSDP.

QUORUM

- No business may be conducted without a quorum.

- The meeting will be disbanded if a quorum is not present within 30 minutes of starting time and reconvened at a time decided by those present.
- In the event that a quorum has not been achieved, the Chairperson shall reschedule the meetings to a later date and a suitable date and time, within ten (10) working days to discuss the same agenda.

RESOLUTIONS

- The Technical Working Group has a duty to make recommendations on matters before it and should not defer unless further information is required.
- Recommendations are made by consensus where possible.

POWERS OF THE WORKING GROUP

The Technical Working Group is an advisory only to the NDoH SEM and, as such, has no authority to make commitments, either financial or contractual on behalf of the NDoH. All decisions will take the form of a recommendation for ratification, amendment or rejection.

CONFLICT OF INTEREST, PROBITY AND CONFIDENTIALITY

All members of the Technical Working Group are required to observe the highest ethical standards, to observe strict confidentiality and impartiality with respect to the business of the Working Group and to immediately declare any conflict of interest prior to the deliberations of the Working Group. Members may receive information that is regarded as commercial in confidence, clinically confidential or have privacy implications. Members acknowledge their responsibilities to maintain confidentiality of all information that is not in the public domain.

A conflict of interest may occur when a member, a member of that member's family or a business associate has an interest in a matter to be discussed. In such circumstances that member should seek leave from the meeting and take no part in the deliberations with respect to that matter.

REPORTING

The NHSS Technical Working Group minutes of meetings forms part of NDoH National Health Standards Services meeting papers and relevant briefings will be provided to the NDOH Deputy Secretaries in order that senior members of the SEM are fully informed of the Working Group's deliberations.

TIME LIMITED TECHNICAL WORKING GROUP

The Technical Working Group is time limited to meet the planned activities and policy framework required in the development of the Review and Development of the PNG NHSS 2012-2030.



Dr Osborne Liko
Acting Executive Manager, Chief Surgeon,
Medical Standards Division

INITIAL NHSS TWG TORS PREVIOUSLY
ENDORSED

Dated: 22nd May 2020.

DEFINITIONS

ACCREDITATION

A recognized body (government) assesses and then recognises that a health care facility meets pre-established performance standards. A committee of experts work with the accrediting body develop the standards for accreditation. They periodically revise the standards to reflect advances in technology, treatment regimes, or policy changes. To conduct accreditation of a health care facility external surveyors evaluate the facility's achievements in providing services and goods to clients/ patients. The survey team then recommends whether the facility should be accredited or should implement further improvements and be re-evaluated. Renewal of accreditation status is usually required every 2-3 years.

AUDIT AND FEEDBACK

Audit and feedback processes encompass a wide variety of interventions, including performance review, supervisor assessment, medical record review, self-audit, self-assessment and peer review. Using structured checklists, or review of records based on best practice evidence.

DEVELOPING GUIDELINES

Identifying and refining the subject area/prioritizing topics, convening guideline development group, systemic reviews, evidence into a recommendation within a clinical practice guideline.

REORGANISING WORK BETTER

Decentralization and integration of health services and increasing the push to reach the poorest and most remote communities is a challenge in all health care organizations. The organization of work approach can help i.e. resources and processes working together – to work more efficiently and effectively:

- Use of evidence-based practices: applying guidance based on demonstrated impacts and eliminating unnecessary procedure barriers
- Adaptability: being flexible to meet changing conditions common in health care service delivery
- Links with services & sites: improve internal and external referral systems
- Minimizing paperwork and maximizing information use: collecting, recording, communicating, applying the right information effectively
- Physical factors: encouraging staff to be resourceful when using facility space and supplies
- Service hours and scheduling: tailoring clinical hours, scheduling, and follow up visits to meet both client and provider needs
- Client flow: shortening wait times, balance client load and client flow
- Division of labour and job design: defining staff responsibilities and functions, lines of authority and structures
- Social factors: providing leadership, motivation, encouragement skills development and positive human relations.

COLLABORATIVE IMPROVEMENTS – POSSIBLE CLINICAL NETWORKS

- A structured improvement approach that organises a large number of teams or sites to work together to achieve a significant improvement in specific areas of care – teamwork, processes, standards, indicators, training, coaching.

ANNEX 2:

HEALTH INFORMATION: HEALTH SECTOR PERFORMANCE INDICATORS FOR HEALTH SERVICES

2.1 ENHIS INDICATORS

	Ind #	Indicator
Outcome	1	Case fatality rate (in HC & Hospitals) for pneumonia in children <5yrs
Outcome	2	Proportion (%) of underweight children under five years
Outcome	3	Underweight (<2500 gm) births as a proportion (%) of total births
Outcome	4	Incidence (0/00) of malaria
Outcome	5	Proportion (%) of pregnant 15 – 24 year old women who test HIV positive
Outcome	6	Incidence (0/00) of diarrhoeal disease in children under 5 years
Outcome	7	Total injury discharges from health centres & hospitals for every 1000 pop'n
Output	8	Ratio of rural outreach clinics held to children under 5 years;
Output	9a	Proportion (%) of children < 1yr who are immunised against measles
Output	9b	Proportion (%) of children < 1yr who are immunised with 3 doses TA/Pentavalent vaccine
Output	10	Proportion (%) of births attended by skilled personnel at health facilities;
Output	11	Proportion (%) of pregnant women who attended at least one ANC visit.
Output	12	Family Planning: couple years protection per thousand WRA
Output	16	Case detection rate of all TB cases
Output	17	TB treatment success rate
Process	18	Proportion (%) of funds that are expended.
Process	19	Provincial health expenditure as a % of required minimum health expenditure.
Process	20	Proportion (%) of health centres that have received at least one supervisory visit
Process	21	Average number of outpatient visits to hospitals & health centres per person per year
Input	24	Total budget allocation (HSIP & GoPNG) per capita
Input	26	Proportion (%) of health centres/hospitals with functioning radio/telephone/mobile.
Input	27	% of months that facilities do not have stock-outs of selected supplies.
Input	28	Proportion (%) of general hospitals (PMGH & the provincial hospitals) which have at least 3 of the 5 key specialties

2.2 HEALTH SECTOR PERFORMANCE INDICATORS FOR HEALTH SERVICES

SEE SEPARATE DOCUMENT

ANNEX 3:

NHSS VOL 1.1 NHSS LEVELS OF CARE AND HEALTH SERVICES MATRIX

SEE SEPARATE DOCUMENT

ANNEX 4:

**NHSS VOL 1.2 NHSS CLINICAL PUBLIC HEALTH GOVERNANCE
FRAMEWORK AND MANUAL**

SEE SEPARATE DOCUMENT

ANNEX 5:

**NHSS VOL 1.3 NATIONAL PATIENT REFERRAL GUIDELINE FOR HEALTH
SERVICES IN PNG**

SEE SEPARATE DOCUMENT

ANNEX 6:

LIST OF NATIONAL DEPARTMENT OF HEALTH CLINICAL GUIDELINES / TREATMENT MANUALS

Area	Type	Name	Title
Adult	Treatment Standard	Standard of Patient Care Treatment for Adult	Standard Treatment for Common Illnesses of Adults in Papua New Guinea: A Manual for Nurses, Health Extension Officers and
Anaesthetics	Guideline	Anaesthetic Guide for PNG	Anaesthetic Guide for PNG
Emergency Medicine	Procedure Manual	Accident and Emergency Procedure Manual	Accident and Emergency Procedure and Practice Manual
ENT	Guideline		ENT Surgical Atlas
Family Planning	Training Manual	Violence and Gender Issues? - Training Manual	Gender Equity - Training Manual
Family Planning	Treatment Standard	Manual of Family Planning for Doctors, HEOs and Nurses in PNG	Manual of Family Planning for Doctors, HEOs and Nurses in PNG (2nd Edition)
Family Planning	Guideline	WHO Family Planning Manual	WHO Family Planning: A Global Handbook for Providers
Infection control	Guideline	Infection Prevention Policy Guidelines for Health Facilities	Infection Prevention Policy Guidelines for Health Facilities (2nd Edition)
Intensive Care	Procedure Manual	ICU Procedure Manual	ICU Procedure Manual
Mental Health	Information	Community Mental Health Information Packages	
Mental Health	Guideline	Mental Health Care for PNG	Mental Health Care for PNG
Nursing	Procedure manual	Nurses Standard Procedure Manual	Nurses Standard Procedure Manual
Nursing	Standard		Papua New Guinea Nursing Competency Standards
Nursing	Standard	Standard of Patient Care	Standards of Patient Care (2nd Edition)
Obstetrics and Gynaecology	Training manual	Emergency Obstetrics Care Training Manual	Emergency Obstetrics Care Training Manual
Obstetrics and Gynaecology	Treatment Standard	O&G Standard Treatment Manual	Manual of Standard Managements in Obstetrics and Gynaecology for Doctors, HEOs and Nurses in Papua New Guinea
Oncology	Guideline	Cancer Management Cancer Review Surgical	Cancer Management Cancer Review Surgical
Oncology	Guideline	Guidelines for the Treatment of Cancer in Papua New Guinea (3rd Edition)	Guidelines for the Treatment of Cancer in Papua New Guinea (3rd Edition)
Ophthalmology	Procedure Manual	Eye Care Standard Procedure Manual	Eye Care Standard Procedure Manual
Ophthalmology	Guideline	Primary Eye Care: A Simple Guide	Primary Eye Care: A Simple Guide
Oral Health	Procedure Manual	Oral Care Curative Health Procedure	Oral Health Services Procedure Manual: For The Oral and Other Health Workers in Papua New Guinea (1st Edition)
Oral Health	Treatment Standard	Oral Health Services: Minimum Standards Manual (1st Edition)	Oral Health Services: Minimum Standards Manual (1st Edition)
Paediatrics	Guideline	Paediatrics for Doctors in Papua New Guinea	Paediatrics for Doctors in Papua New Guinea (2nd Edition)
Paediatrics	Treatment Standard	Children Standard Treatment	Standard Treatment for Common Illnesses of Children in Papua New Guinea: A Manual for Nurses, Community Health Workers,
Pathology	Procedure Manual		General Laboratory Manual
Pathology	Procedure Manual		Media
Pathology	Procedure Manual		Laboratory Procedure Manual
Public Health	Guideline		Waste Management
Radiography	Procedure Manual	Radiographers Procedure Manual	Radiography Procedure and Practice Manual
Service Delivery	Standard		Minimum Standards for District Health Services in PNG
Service Delivery	Standard		Priorities, Policies, and Standards of Curative Health Services as per National Health Plan 2001-2010
STI	Standard	HIV Prophylaxis - WHO	Standard Management of Sexually Transmitted Infections and Genital Conditions in Papua New Guinea: A Manual for Health
STI	Standard		Standard Management of Sexually Transmitted Infections (STIs) and Genital Conditions in Papua New Guinea: A Manual for STI
STI	Standard	STI Clinic Standards?	Minimum Standards for STI Service Delivery in Papua New Guinea A reference manual for all health care providers in PNG
Surgery	Hand book	HEO and Nurses Surgical Hand Book	HEO and Nurses Surgical Hand Book
Surgery	Procedure Manual	Operating Theatre Management Procedure Manual	Operating Theatre Procedure and Practice Manual
Surgery	Guideline	Standard Tray for Operating Theatre	Standard Tray for Operating Theatre
Surgery	Guideline	Surgery: For Primary Health Care Workers in Papua New Guinea	Surgery: For Primary Health Care Workers in Papua New Guinea
Traditional Medicine	Policy	Traditional Medicine Policy Formulation Conference Proceedings	Traditional Medicine Policy